PhD Thesis — Pilot Study

Artificial Intelligence for Balance of Payments Estimation: Nowcasting, Alternative Data, and Policy Implications

Toma Savitki

[University / Institution]

[Doctoral School]

Supervisor: [Supervisor Name]

March 2026

First draft — pilot study version

Abstract

This thesis investigates whether artificial intelligence and machine learning methods can improve the timeliness and accuracy of France's Balance of Payments (BoP) estimates, and whether such improvements carry policy relevance. Three self-contained chapters address this question.

Chapter 1 conducts a systematic comparison of nine models—spanning autoregressive benchmarks, econometric approaches (Dynamic Factor Model, Bridge equation), regularised linear machine learning (Ridge regression), nonlinear ensembles (Gradient Boosting, XGBoost), and deep learning (LSTM, GRU)—for nowcasting France's quarterly current account balance over 2003–2022. In a pseudo-real-time expanding-window evaluation, Ridge regression achieves the largest and most robust improvement: a 26.3% RMSE reduction over AR(1), significant at the 5% level via the Diebold–Mariano test. Both LSTM and GRU fail entirely, confirming that deep learning is ill-suited to small-sample macroeconomic settings.

Chapter 2 conducts a structured ablation study to quantify the incremental value of four categories of alternative data—trade-activity proxies, financial-flow indicators, text/sentiment data, and satellite/geospatial proxies—for monthly goods trade balance nowcasting. The full model incorporating all categories achieves a 6.4% RMSE reduction over the baseline (DM $p = 0.060$; GW $p = 0.195$), corroborated by a pseudo-real-time vintage robustness check showing a 6.0% gain across 50 Monte Carlo replications. SHAP decomposition reveals that text/sentiment data provide the largest marginal contribution (19.0% of total importance), followed by trade-activity proxies (8.5%).

Chapter 3 bridges the gap between statistical accuracy and policy relevance through three exercises. Crisis evaluation shows that Ridge nowcasts maintained substantial accuracy gains during COVID-19 (-17.7% RMSE) and the energy shock (-19.8%). Cross-country tests confirm generalisability to France, Germany, and Italy (-16 to -20% RMSE, all $p < 0.01$), while Spain shows a negligible improvement (-2.5%, $p = 0.559$). A BoP-augmented Taylor rule yields a long-run CA gap coefficient of -0.46; IV and Prais–Winsten GLS robustness checks confirm that neither endogeneity nor serial correlation biases the conclusions.

Keywords: Balance of Payments, nowcasting, machine learning, Ridge regression,

alternative data, NLP sentiment, SHAP, Taylor rule, current account

JEL Classification: C53, C55, E37, F32, F47

Résumé

Cette thèse examine si les méthodes d'intelligence artificielle et d'apprentissage automatique peuvent améliorer la rapidité et la précision des estimations de la Balance des Paiements (BdP) de la France, et si ces améliorations sont pertinentes pour la politique économique. Trois chapitres autonomes abordent cette question.

Le Chapitre 1 réalise une comparaison systématique de neuf modèles pour le *nowcasting* du solde courant trimestriel français (2003–2022). La régression Ridge obtient la meilleure amélioration : une réduction du RMSE de 26,3 % par rapport à l'AR(1), significative au seuil de 5 % (DM $p = 0{,}013$).

Le Chapitre 2 conduit une étude d'ablation structurée quantifiant la valeur incrémentale de quatre catégories de données alternatives. Le modèle complet réalise une réduction de 6,4 % du RMSE (DM $p = 0{,}060$), les données textuelles/sentiment fournissant la plus grande contribution marginale (19,0 %).

Le Chapitre 3 comble l'écart entre précision statistique et pertinence politique à travers une évaluation en période de crise, des tests de généralisabilité transnationaux, et une analyse contrefactuelle de politique monétaire. Une règle de Taylor augmentée avec lissage des taux d'intérêt produit un coefficient de long terme du solde courant de $-0{,}46$; l'écart maximal entre les taux directeurs implicites calculés avec le *nowcast* et avec les données officielles atteint 2,6 points de base.

Mots-clés : Balance des paiements, nowcasting, apprentissage automatique, régression Ridge, données alternatives, sentiment NLP, SHAP, règle de Taylor, compte courant

Acknowledgements

I would like to express my sincere gratitude to my thesis supervisor, [Supervisor Name], for guidance, patience, and constructive feedback throughout this research project.

I thank [Institution/University] and the doctoral school for providing the academic environment and resources that made this work possible.

I am grateful to the Banque de France, the European Central Bank, and the various public data providers (FRED, Eurostat, Google Trends) whose open-access data policies made this empirical research feasible.

Any remaining errors are my own.

Paris, March 2026

Contents

Abstract **2**

Résumé **4**

Acknowledgements **i**

General Introduction **1**

1 AI vs. Econometrics for BoP Nowcasting: Which Models Work Best? **6**

1.1 Introduction 2
1.1.1 Survey of Nowcasting Methods 2
1.2 Data 4
1.2.1 Target Variable 4
1.2.2 Predictor Variables 4
1.2.3 Unit-Root Diagnostics 5
1.3 Methodology 5
1.3.1 Models 5
1.3.2 Evaluation Protocol 7
1.3.3 Statistical Testing 7
1.4 Results 7
1.4.1 Quarterly Current Account 7
1.4.2 Monthly Goods Trade Balance 10
1.4.3 Bootstrap Confidence Intervals 10
1.4.4 Sub-Period Stability 11
1.4.5 Hyperparameter Sensitivity 11
1.4.6 Clark–West Nested-Model Tests 12
1.4.7 Multiple-Testing Correction 13
1.4.8 Forecast Combination 13
1.4.9 Model Confidence Set 14
1.5 Discussion 14
1.5.1 Real-Time Information Flow Test 15

1.5.2 Feature Importance . . . 15
1.6 Conclusion . . . 17

2 The Value of Alternative Data: Which Sources Help? 18
2.1 Introduction . . . 2
2.1.1 Contribution . . . 2
2.2 Data . . . 3
2.2.1 Target . . . 3
2.2.2 Baseline Predictors (Category B) . . . 3
2.2.3 Alternative Data . . . 3
2.3 Methodology . . . 4
2.3.1 Ablation Design . . . 4
2.3.2 Evaluation Protocol . . . 4
2.3.3 Statistical Tests . . . 4
2.3.4 SHAP Decomposition . . . 5
2.4 Results . . . 5
2.4.1 Ablation Results . . . 5
2.4.2 Bootstrap Confidence Intervals . . . 6
2.4.3 Model Confidence Set . . . 6
2.4.4 Giacomini–Rossi Fluctuation Test . . . 6
2.4.5 M5 Interaction Analysis . . . 7
2.4.6 Out-of-Sample Stability Tests . . . 7
2.4.7 SHAP Decomposition . . . 8
2.5 Discussion . . . 9
2.5.1 Why Text/Sentiment Data Lead . . . 9
2.5.2 The Aggregation Premium . . . 10
2.5.3 Economic vs. Statistical Significance . . . 10
2.5.4 Cost–Benefit Analysis . . . 10
2.6 Robustness . . . 11
2.6.1 Rolling Window . . . 11
2.6.2 Alternative ML Methods . . . 11
2.6.3 Pseudo-Real-Time Vintage Robustness . . . 11
2.7 Conclusion . . . 11
.1 NLP Pipeline Details . . . 12
.1.1 ECB Statement Processing . . . 12
.1.2 BdF Business Climate Indicator . . . 13
.2 Satellite Data . . . 13

A Policy Implications: Do Better Estimates Matter? **14**
A.1 Introduction . . . 2
A.2 Part A: Crisis Episode Evaluation . . . 2
A.2.1 Design . . . 2
A.2.2 Results . . . 3
A.2.3 Interpretation . . . 4
A.3 Part B: Cross-Country Generalisability . . . 4
A.3.1 Design . . . 4
A.3.2 Results . . . 4
A.3.3 Transfer Learning . . . 5
A.4 Part C: Counterfactual Policy Analysis . . . 6
A.4.1 A BoP-Augmented Taylor Rule . . . 6
A.4.2 Estimation Results . . . 6
A.4.3 Counterfactual Exercise . . . 8
A.4.4 Asymmetric Response Test . . . 8
A.4.5 Institutional Comparison . . . 9
A.4.6 Svensson Optimal-Policy Loss . . . 10
A.5 Discussion . . . 10
A.6 Conclusion . . . 12

General Conclusion **13**

A Data Dictionary and Model Parameters **17**
A.1 Complete Variable Dictionary . . . 17
A.2 Model Hyperparameters . . . 17
A.3 Bootstrap Details . . . 17
A.3.1 Block Bootstrap Procedure . . . 17
A.3.2 Block Length Selection . . . 20
A.4 NLP Pipeline Technical Details . . . 20
A.4.1 FinBERT Configuration . . . 20
A.4.2 ECB Press Conference Processing . . . 20
A.4.3 Additional Text Sources . . . 21
A.5 Satellite Data Construction . . . 21
A.6 Reproducibility . . . 21
A.6.1 Software Environment . . . 21
A.6.2 Data Access . . . 22
A.6.3 Code Availability . . . 22

References **23**

List of Tables

1.1 Unit-root diagnostics: ADF and KPSS test results 5
1.2 Model comparison: Quarterly current account balance (France, 2010–2022) 7
1.3 Bootstrap 95% confidence intervals for RMSE improvement vs. AR(1) 10
1.4 Sub-period RMSE and % improvement vs. AR(1) by model 11
1.5 Hyperparameter sensitivity: RMSE under alternative configurations . 12
1.6 Clark–West test and out-of-sample R^2 vs. AR(1) 12
1.7 Holm–Bonferroni adjusted p-values for DM and CW tests 13

2.1 Alternative data categories and sources 3
2.2 Ablation model specifications 4
2.3 Ablation study: RMSE and improvement over M0 baseline 5
2.4 Bootstrap 95% confidence intervals for RMSE improvement vs. M0 . 6
2.5 SHAP importance decomposition by data category (M5 Full model) . 8
2.6 Cost–benefit assessment of alternative data categories 10

A.1 Crisis episodes 2
A.2 Crisis episode evaluation: RMSE and improvement vs. AR(1) 3
A.3 Crisis significance tests: Ridge vs. AR(1) 4
A.4 Cross-country generalisability: Ridge RMSE improvement vs. AR(1) . 4
A.5 BoP-augmented Taylor rule with interest-rate smoothing (2003–2022) 6

A.1 Complete variable dictionary 18
A.2 Model hyperparameters 19

List of Figures

1.1 RMSE comparison across nine models for quarterly current account nowcasting (France, 2010–2022). Lower bars indicate better performance. Ridge regression achieves the lowest RMSE. 8

1.2 Actual vs. predicted quarterly current account balance (France, 2010–2022) for AR(1), Ridge, and XGBoost. 9

1.3 Cumulative squared forecast errors over the evaluation window. Ridge accumulates the least error; LSTM the most. 9

1.4 Distribution of one-step-ahead forecast errors by model. Ridge (green) is most tightly centred on zero. 10

1.5 Mean absolute SHAP feature importance for the Ridge model. 16

1.6 SHAP summary plot: each dot represents one observation; colour indicates feature value (red = high, blue = low). 16

2.1 Giacomini–Rossi fluctuation test: rolling DM statistic for M5 Ridge vs. M0 Ridge. The dashed lines mark the 5% critical values. M5 is significantly better than M0 during 2.4% of the evaluation window. . 7

2.2 SHAP importance decomposition by data category for the M5 (Full) model. Conventional macro predictors account for 59% of forecast variation; text/sentiment data provide the largest alternative data contribution (19%). 8

2.3 Feature-level SHAP summary plot for the M5 model. Each dot represents one evaluation date; colour indicates feature value (red = high, blue = low). 9

A.1 RMSE comparison across crisis episodes (GFC, COVID, Energy shock) for selected models vs. AR(1). 3

A.2 Ridge regression RMSE improvement vs. AR(1) across four Euro-area economies. Improvements for France, Germany, and Italy are statistically significant at the 5% level; Spain's improvement is not significant ($p = 0.559$). 5

A.3 Counterfactual implied policy rates: official BoP data vs. Ridge nowcast. 9

General Introduction

Context and Motivation

The Balance of Payments (BoP) is one of the most important macroeconomic statistical frameworks. It records all economic transactions between residents of a country and the rest of the world, covering trade in goods and services, primary and secondary income flows, and financial account transactions. BoP statistics directly inform major policy decisions: the European Commission's Macroeconomic Imbalance Procedure (MIP) uses current account balances to trigger corrective action; central banks monitor external positions for financial stability risks; and trade negotiators rely on BoP data to assess the impact of commercial policy.

Yet BoP statistics suffer from a fundamental tension between their policy importance and their timeliness. For France—the focus of this thesis—the quarterly current account balance is published by the Banque de France approximately three months after the end of the reference quarter (International Monetary Fund, 2009). Monthly goods trade data are available with a shorter delay (approximately six weeks), but they capture only one component of the current account. During this publication gap, policymakers must rely on informal estimates, partial indicators, or simple extrapolation.

This tension has grown more acute in recent years for three reasons.

First, the frequency and severity of external shocks has increased. The 2008–09 Global Financial Crisis produced sudden stops in capital flows and rapid current account reversals. The 2020 COVID-19 pandemic caused an unprecedented collapse in trade volumes within weeks. The 2022 energy price shock triggered a dramatic deterioration of the Euro-area terms of trade. In each episode, timely BoP information would have been valuable for calibrating policy responses—but was unavailable when most needed.

Second, the proliferation of high-frequency and alternative data sources creates new opportunities for real-time economic monitoring. Satellite imagery of port congestion, daily vessel tracking data, real-time payment flows, Google search patterns, and natural language processing of central bank communications can all potentially provide early signals about external sector developments. Whether these data sources

actually improve BoP estimation is an empirical question that remains unanswered.

Third, advances in machine learning and artificial intelligence have transformed forecasting in many domains—from weather prediction to retail demand—but their application to macroeconomic statistics remains in its infancy, particularly for external sector data. While a rich literature has emerged on ML for GDP nowcasting (Bańbura et al., 2013; Giannone et al., 2008; Goulet Coulombe et al., 2022), there is no comparable body of work for BoP estimation.

This thesis addresses all three challenges. It develops and evaluates AI-driven methods to improve the timeliness and accuracy of France's Balance of Payments estimates, using both traditional macroeconomic indicators and alternative data sources, and assesses whether the resulting improvements matter for policy.

Literature Positioning

This thesis sits at the intersection of three research strands.

Macroeconomic Nowcasting

The modern nowcasting literature begins with Giannone et al. (2008), who introduced dynamic factor models for GDP estimation using mixed-frequency data. Bańbura et al. (2013, 2010) extended this framework to large Bayesian VARs and provided a comprehensive survey of the now-casting methodology. The key insight is that timely high-frequency indicators (industrial production, surveys, financial variables) contain information about low-frequency aggregates that can be extracted in real time.

Bok et al. (2018) survey the state of the art, documenting how central banks worldwide have operationalised nowcasting. The Federal Reserve Bank of New York's "Nowcast", the ECB's "Euro Area Real-Time GDP" model, and the Bank of England's suite of nowcasting tools all use variants of this framework. Doz et al. (2012) develop quasi-maximum likelihood methods for large approximate dynamic factor models, while Mariano and Murasawa (2003) and Camacho and Perez-Quiros (2010) introduce coincident indicators and short-term prediction models for the Euro area.

However, this literature has focused almost exclusively on GDP and its components (consumption, investment). External sector statistics—current account, trade balance, financial flows—have received comparatively little attention, despite their importance for policy surveillance (Obstfeld, 2012). This thesis fills this gap.

Machine Learning for Economic Forecasting

A second literature brings machine learning methods to macroeconomic forecasting. Goulet Coulombe et al. (2022) provide a definitive assessment, finding that regularised linear models (Ridge, LASSO) often dominate more complex methods in low-dimensional macroeconomic settings. Babii et al. (2022) develop theoretical foundations for sparse ML time series regressions. Medeiros et al. (2021) apply similar methods to inflation forecasting. Athey and Imbens (2019) survey the broader landscape of ML methods for economics, and H. R. Varian (2014) discusses the opportunities created by big data for econometric practice.

Key findings from this literature that inform our approach: (i) regularisation is essential when sample sizes are small relative to feature dimensions (Hoerl and Kennard, 1970; Tibshirani, 1996); (ii) tree-based ensembles (XGBoost, Random Forest) can capture nonlinearities but tend to overfit with fewer than several hundred observations (Chen and Guestrin, 2016); (iii) deep learning methods (LSTM, Transformer) generally require much larger samples than macroeconomic settings provide, unless pre-training or transfer learning is employed (Hochreiter and Schmidhuber, 1997; Lim et al., 2021).

Alternative Data for Economics

A rapidly growing literature explores non-traditional data sources for economic monitoring. Woloszko (2020) uses Google Trends for real-time activity tracking at the OECD. Aprigliano et al. (2019) use payment system data to forecast Italian GDP. Caldara and Iacoviello (2022) construct geopolitical risk indices from news text. In the financial domain, FinBERT (Araci, 2019) and similar NLP models extract sentiment from financial communications (S. Hansen and McMahon, 2016; Shapiro et al., 2022). Ferrara and Simoni (2023) examine when Google data are useful for GDP nowcasting via pre-selection and shrinkage. Choi and H. Varian (2012) demonstrate early applications of Google Trends for economic prediction.

For external sector data specifically, the Baltic Dry Index has long been used as a leading indicator for trade, and recent work with AIS vessel tracking and satellite imagery of ports suggests these sources could provide very early signals of trade disruptions (Donaldson and Storeygard, 2016). The contribution of this thesis is to systematically quantify the incremental value of these data categories for BoP estimation.

Research Questions and Thesis Structure

The thesis is organised as three self-contained essays (*thèse sur articles* format), each addressing a distinct research question:

Chapter 1: AI vs. Econometrics for BoP Nowcasting — Which Models Work Best? The first chapter conducts a systematic horse race between econometric and machine learning approaches for BoP nowcasting. Using quarterly current account and monthly goods trade balance data for France (2000–2022), it compares nine models: autoregressive benchmarks AR(1) and AR(4), a dynamic factor model (Stock and Watson, 2002), a bridge equation (Baffigi et al., 2004), Ridge regression, Gradient Boosting, XGBoost, LSTM, and GRU. In a pseudo real-time expanding-window evaluation, Ridge regression delivers the most robust improvement: a 26.3% RMSE reduction over AR(1) at quarterly frequency (statistically significant at 5% via Diebold and Mariano 1995 test). Both LSTM and GRU fail entirely, confirming that deep learning needs larger samples.

Chapter 2: The Value of Alternative Data — Which Data Sources Help? Building on Chapter 1's Ridge baseline, the second chapter conducts a structured ablation study to quantify the incremental value of four categories of alternative data: trade-activity proxies, financial-flow indicators, text/sentiment data, and satellite/geospatial proxies. Using monthly data for France (2008–2022), the full model incorporating all categories achieves a 6.4% RMSE reduction over baseline Ridge (DM $p = 0.060$), with text and sentiment data providing the largest SHAP contribution (19.0%). All data sources are from real public APIs, though ECB text coverage and satellite proxies are partially limited (see Chapter 2 discussion).

Chapter 3: Policy Implications — Do Better Estimates Matter? The third chapter bridges the gap between statistical accuracy and policy relevance. It evaluates nowcast performance during three crisis episodes (GFC, COVID, 2022 energy shock), tests cross-country generalisability across four Eurozone economies (FR, DE, IT, ES), and conducts a counterfactual policy exercise using a Taylor-type reaction function augmented with BoP signals. A BoP-augmented Taylor rule with interest-rate smoothing yields a long-run current account gap coefficient of -0.46; the maximum gap between nowcast-implied and official-data-implied policy rates reaches 2.6 basis points at the monthly level, concentrated during crisis episodes.

Data

The thesis uses two principal data sources:

- **Official statistics:** Quarterly and monthly BoP components from the ECB Statistical Data Warehouse and Eurostat (publicly available). Standard macroeconomic indicators—GDP, industrial production, HICP, exchange rates, short-term interest rates—are obtained from the same source.
- **Alternative data:** Eight real data sources from public APIs (FRED, Google Trends, Caldara–Iacoviello website), including the Baltic Dry Index, OECD container throughput, EUR/USD volatility, France–Germany bond spread, ECB FinBERT sentiment, Banque de France Business Climate Indicator, geopolitical risk index, and nighttime light intensity.

Methodology

All empirical analyses use the expanding-window pseudo real-time evaluation protocol with Diebold and Mariano (1995) tests for statistical significance and Lundberg and Lee (2017) SHAP values for model interpretability. Bootstrap confidence intervals (Efron, 1979; Künsch, 1989) provide uncertainty quantification for RMSE improvement estimates. The evaluation protocol is described in detail in each chapter.

Chapter 1

AI vs. Econometrics for BoP Nowcasting: Which Models Work Best?

Abstract

This chapter conducts a systematic comparison of econometric and machine-learning approaches for nowcasting France's quarterly current account balance. Nine models—AR(1), AR(4), Dynamic Factor Model, Bridge equation, Ridge regression, Gradient Boosting, XGBoost, LSTM, and GRU—are evaluated on both quarterly current account and monthly goods trade data over 2003–2022 using an expanding-window protocol with a minimum training sample of 40 quarters (10 years). Ridge regression delivers the largest and most robust improvement: a 26.3% RMSE reduction over AR(1) at the quarterly frequency, significant at the 5% level via the Diebold and Mariano (1995) test ($p = 0.013$) and at the 1% level via the Clark and West (2007) nested-model test ($p < 0.001$), with an out-of-sample R^2 of 0.46. Traditional econometric approaches (DFM, Bridge) offer moderate but less consistent gains. Complex nonlinear methods—Gradient Boosting, XGBoost—improve on the benchmark but are outperformed by Ridge. Both LSTM (+1.0% RMSE) and GRU (+1.0%) fail entirely, confirming that deep learning—regardless of recurrent architecture—is ill-suited to the small-sample macroeconomic setting. These results corroborate Goulet Coulombe et al. (2022)'s finding that regularised linear models dominate in low-dimensional macroeconomic forecasting.

1.1 Introduction

Timely monitoring of external sector developments is critical for macroeconomic policymaking. The Balance of Payments (BoP) records all transactions between a country's residents and the rest of the world, and the current account balance—the broadest summary measure of these flows—directly enters the European Commission's Macroeconomic Imbalance Procedure (MIP) and influences monetary policy deliberations (European Commission, 2016; Obstfeld, 2012).

For France, the quarterly current account balance is published by the Banque de France with a delay of approximately three months. Monthly goods trade data (from Customs) are available sooner but cover only one component. This publication gap creates a "nowcasting" problem: how to produce the best possible estimate of the current or most recent quarter's BoP given the information available today.

A rich literature addresses GDP nowcasting using dynamic factor models (Giannone et al., 2008; Stock and Watson, 2002), bridge equations (Baffigi et al., 2004; Foroni et al., 2015), and—more recently—machine learning methods (Babii et al., 2022; Goulet Coulombe et al., 2022). Yet external sector statistics have received little attention. Ghysels and Ozkan (2015) discuss the potential of mixed-frequency methods for BoP data, and Ca' Zorzi et al. (2017) examine exchange-rate models for current account prediction but do not conduct real-time evaluation.

This chapter fills the gap by providing the first systematic model comparison for BoP nowcasting. Our contributions are threefold:

1. We benchmark nine models spanning four methodological families (autoregressive, factor/bridge econometric, regularised linear ML, nonlinear ML) on the same data and evaluation protocol.

2. We use an expanding-window design that mimics real-time data availability, with formal statistical tests (Diebold and Mariano, 1995) for forecast accuracy comparisons.

3. We document a clear "regularisation premium": Ridge regression dominates all alternatives, while deep learning (LSTM) underperforms the simplest benchmark.

1.1.1 Survey of Nowcasting Methods

Dynamic Factor Models. The seminal work of Stock and Watson (2002) established the proposition that a few latent factors can summarise the information in large macroeconomic panels. Giannone et al. (2008) operationalised this insight for real-time GDP tracking, and Bańbura et al. (2010) extended the framework to

handle hundreds of series through Bayesian shrinkage. Doz et al. (2012) developed a quasi-maximum likelihood estimator that is robust to cross-sectional correlation in the idiosyncratic component. DFMs remain the workhorse of institutional nowcasting: the ECB, Federal Reserve Bank of New York, and Banca d'Italia all run DFM-based nowcasting suites (Bok et al., 2018). However, DFMs assume linear factor structure and may miss nonlinear interactions.

Bridge Equations and MIDAS. Bridge equations link monthly indicators to quarterly targets via temporal aggregation constraints (Baffigi et al., 2004). Foroni et al. (2015) propose unrestricted MIDAS (U-MIDAS) regressions that avoid the restrictive polynomial lag structures of standard MIDAS while preserving the mixed-frequency advantage. Mariano and Murasawa (2003) introduce business-cycle coincident indicators that can serve as bridge variables. Camacho and Perez-Quiros (2010) implement Markov-switching bridge equations for Euro-area short-term prediction. Bridge models are parsimonious and interpretable but depend critically on the choice of bridge variable and may fail when the leading relationship between the monthly indicator and the quarterly target breaks down.

Machine Learning Methods. Goulet Coulombe et al. (2022) provide the most comprehensive assessment of ML for macroeconomic forecasting to date. Using a large panel of US macroeconomic series, they find that Ridge regression—a simple ℓ_2-penalised OLS estimator (Hoerl and Kennard, 1970)—matches or outperforms more complex methods including LASSO (Tibshirani, 1996), Random Forest, Gradient Boosting, and neural networks. The key mechanism is bias–variance tradeoff: in small samples with moderate dimensionality, aggressive regularisation reduces variance more than it increases bias.

Medeiros et al. (2021) reach similar conclusions for inflation forecasting in Brazil: tree-based methods and Ridge both outperform benchmark autoregressions, but Ridge does so more consistently. Babii et al. (2022) develop the theoretical underpinnings for machine-learning time-series regressions, establishing oracle inequalities under β-mixing conditions. Athey and Imbens (2019) provide a broader perspective on how ML tools should be adapted for causal and predictive inference in economics, and H. R. Varian (2014) discusses the promise of big data methods for macroeconomic modelling.

For tree-based ensembles, Chen and Guestrin (2016) introduced XGBoost, which has become the standard boosting implementation. These methods excel at capturing nonlinear interactions but are prone to overfitting with fewer than several hundred observations—a binding constraint in quarterly macroeconomic applications.

Deep Learning for Time Series. Hochreiter and Schmidhuber (1997) introduced the Long Short-Term Memory architecture, which can in principle capture long-range dependencies in sequential data. Lim et al. (2021) develop the Temporal Fusion Transformer, which combines attention mechanisms with interpretability features. However, both architectures require sample sizes several orders of magnitude larger than typical macroeconomic datasets provide. Pre-training on large corpora (Devlin et al., 2019) or transfer learning from related domains are promising strategies but remain underexplored for macroeconomic time series.

1.2 Data

1.2.1 Target Variable

We use two target variables for France:

- **Quarterly current account balance** (EUR millions), from the ECB Statistical Data Warehouse, 2000 Q1–2022 Q4 ($T_q = 92$ observations).
- **Monthly goods trade balance** (EUR millions), from the same source, 2000 M1–2022 M12 ($T_m = 276$ observations).

1.2.2 Predictor Variables

The predictor set contains 14 monthly macroeconomic and financial variables:

- **Real activity:** Industrial production index, retail sales, unemployment rate.
- **Survey data:** Manufacturing PMI, consumer confidence.
- **External trade:** Goods exports (value), goods imports (value), terms of trade.
- **Financial:** EUR/USD exchange rate, 3-month Euribor, EURO STOXX 50 returns.
- **Alternative:** Baltic Dry Index, Google Trends ("exportation France"), Banque de France Business Climate Indicator.

All series are seasonally adjusted where applicable, stationarity is ensured through first-differencing or log-differencing, and missing values are interpolated using the EM algorithm following Stock and Watson (2002).

1.2.3 Unit-Root Diagnostics

We assess the stationarity of all variables using both the augmented Dickey–Fuller (ADF) test (Dickey and Fuller, 1979)—whose null hypothesis is a unit root—and the KPSS test (Kwiatkowski et al., 1992)—whose null is stationarity. Combining both tests reduces the risk of erroneous conclusions from either alone. Table 1.1 summarises the results.

Table 1.1: Unit-root diagnostics: ADF and KPSS test results

Variable	ADF stat	ADF p	KPSS stat	KPSS p	Conclusion
bop_ca	−1.95	0.309	1.275	<0.01	Unit root
bop_goods	−0.28	0.928	1.169	<0.01	Unit root
bop_services	−1.69	0.435	1.349	<0.01	Unit root
gdp	−0.26	0.932	1.593	<0.01	Unit root
interest_rate	−1.93	0.320	1.310	<0.01	Unit root
bop_ca_yoy	−7.56	<0.001	0.067	>0.10	Stationary
bop_ca_qoq	−7.96	<0.001	0.101	>0.10	Stationary
gdp_yoy	−3.37	0.012	0.076	>0.10	Stationary
gdp_qoq	−10.73	<0.001	0.083	>0.10	Stationary

Level series (current account, goods trade, GDP, interest rate) exhibit unit roots, while year-on-year and quarter-on-quarter transformations are stationary by both tests. The EUR/USD rate and HICP are classified as inconclusive (ADF fails to reject but KPSS does not reject stationarity either). All non-stationary series enter the models in first-differenced or growth-rate form, ensuring that the forecasting regressions operate on stationary data.

1.3 Methodology

1.3.1 Models

We evaluate nine models spanning four methodological families:

Autoregressive benchmarks.

- **AR(1):** $y_t = c + \phi_1 y_{t-1} + \varepsilon_t$.
- **AR(4):** $y_t = c + \sum_{j=1}^{4} \phi_j y_{t-j} + \varepsilon_t$.

Econometric models.

- **Dynamic Factor Model (DFM):** Extracts k principal components from the standardised predictor panel and uses them in a linear forecasting equation.

The number of factors k is selected automatically at each expanding-window step using the IC_{p2} information criterion of Bai and Ng (2002): $IC_{p2}(k) = \ln V(k) + k\,\frac{N+T}{NT}\,\ln\min(N,T)$, where $V(k)$ is the average squared residual with k factors. Implemented via PCA following Stock and Watson (2002).

- **Bridge equation:** Selects up to five monthly predictors via forward stepwise BIC minimisation at each expanding-window step, then regresses the quarterly target on the quarterly averages of the selected indicators following Baffigi et al. (2004).

Machine Learning – Linear.

- **Ridge regression:** ℓ_2-penalised OLS with regularisation parameter α selected by efficient leave-one-out cross-validation (LOOCV) over the grid $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ using `sklearn.linear_model.RidgeCV` (Hoerl and Kennard, 1970). The LOOCV selector is re-evaluated at each expanding-window step so that α adapts to the growing training sample. All predictors are standardised to zero mean and unit variance before estimation.

Machine Learning – Nonlinear.

- **Gradient Boosting (GB):** Sklearn implementation with hyperparameters selected by three-fold time-series cross-validation (`TimeSeriesSplit`) at each expanding-window step over a grid of learning rate $\in \{0.05, 0.1\}$, max depth $\in \{3, 5\}$, and $n_{\text{estimators}} \in \{100, 200\}$.

- **XGBoost:** Same TSCV protocol as GB with a grid over learning rate $\in \{0.05, 0.1\}$, max depth $\in \{3, 5\}$, and regularisation $\lambda \in \{0, 1\}$ (Chen and Guestrin, 2016).

- **LSTM:** Single-layer LSTM with 32 hidden units, dropout 0.2 between the recurrent and output layers, lookback window of 4 periods, trained with Adam optimiser (lr $= 0.005$). A held-out validation set (last 20% of the training window) is used for early stopping with patience 15 and for a `ReduceLROnPlateau` scheduler (factor 0.5, patience 5). The best validation-loss state is restored before forecasting (Hochreiter and Schmidhuber, 1997).

- **GRU:** Two-layer Gated Recurrent Unit with 32 hidden units, dropout 0.1, weight decay 10^{-4}. Validation-based early stopping (patience 15) and `ReduceLROnPlateau` scheduling follow the same protocol as LSTM (Cho et al., 2014).

1.3.2 Evaluation Protocol

We use an *expanding-window* evaluation design:

1. Fix a minimum training window of 40 quarters (10 years): 2000 Q1–2009 Q4.
2. For each evaluation quarter t from 2010 Q1 to 2022 Q4:
   (a) Estimate the model using data from 2000 Q1 to $t-1$.
   (b) Produce a one-step-ahead forecast $\hat{y}_t$.
3. Compute Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and direction accuracy over the out-of-sample evaluation window.

1.3.3 Statistical Testing

We use the Diebold and Mariano (1995) test with Newey and West (1987) HAC standard errors to assess whether forecast accuracy differences are statistically significant. The null hypothesis is equal predictive ability: $H_0 : E[d_t] = 0$ where $d_t = e_{1,t}^2 - e_{2,t}^2$ is the loss differential between two competing forecasts.

1.4 Results

1.4.1 Quarterly Current Account

Table 1.2 presents the main results for quarterly current account nowcasting.

Table 1.2: Model comparison: Quarterly current account balance (France, 2010–2022)

Model	RMSE	% vs. AR(1)	DM p-value	Direction Acc.
AR(1)	8913	—	—	48%
AR(4)	7512	-15.7	0.114	72%
DFM	7229	-18.9	0.049	80%
Bridge	7685	-13.8	0.288	76%
Ridge	6567	-26.3	0.013	78
Gradient Boosting	7344	-17.6	0.062	72%
XGBoost	7476	-16.1	0.082	72%
LSTM	9004	+1.0	0.669	54%
GRU	9000	+1.0	0.681	42%

Figure 1.1 visualises the RMSE comparison across models.

Several findings emerge.

First, Ridge regression dominates all alternatives, reducing RMSE by 26.3% relative to AR(1). The Diebold and Mariano (1995) test confirms that this improvement

Figure 1.1: RMSE comparison across nine models for quarterly current account nowcasting (France, 2010–2022). Lower bars indicate better performance. Ridge regression achieves the lowest RMSE.

is statistically significant at the 5% level ($p = 0.013$). Ridge also achieves the highest directional accuracy (78%), meaning it correctly predicts whether the current account deteriorates or improves in more than three quarters of the evaluation periods.

Second, traditional econometric approaches provide meaningful improvements. The DFM—with factor count selected by the IC_{p2} criterion—reduces RMSE by 18.9%, now significant at the 5% level ($p = 0.049$), a marked improvement over fixed two-factor specifications and consistent with the importance of data-driven factor selection (Bai and Ng, 2002). The Bridge equation performs moderately (−13.8%), though the DM test does not reject equal accuracy ($p = 0.288$), suggesting that its advantage is less stable across evaluation sub-samples.

Third, nonlinear ML methods improve on the AR(1) benchmark but are dominated by Ridge. Gradient Boosting (−17.6%, $p = 0.062$) and XGBoost (−16.1%, $p = 0.082$) both deliver gains significant at the 10% level but not at the 5% level. Their underperformance relative to Ridge is consistent with the bias–variance tradeoff: with only 51 evaluation points drawn from a 91-quarter sample, the variance reduction from ℓ_2 regularisation dominates the potential nonlinear fit gains from tree ensembles. This finding corroborates Goulet Coulombe et al. (2022).

Fourth, LSTM fails entirely, producing an RMSE 1% *larger* than AR(1). With only 92 quarterly observations and 14 features, the network overfits despite regularisation via early stopping. This failure provides a useful cautionary datum for the literature.

Figure 1.2 presents the time-series of actual vs. predicted current account balances for selected models, and Figure 1.3 shows the cumulative squared forecast errors over the evaluation window.

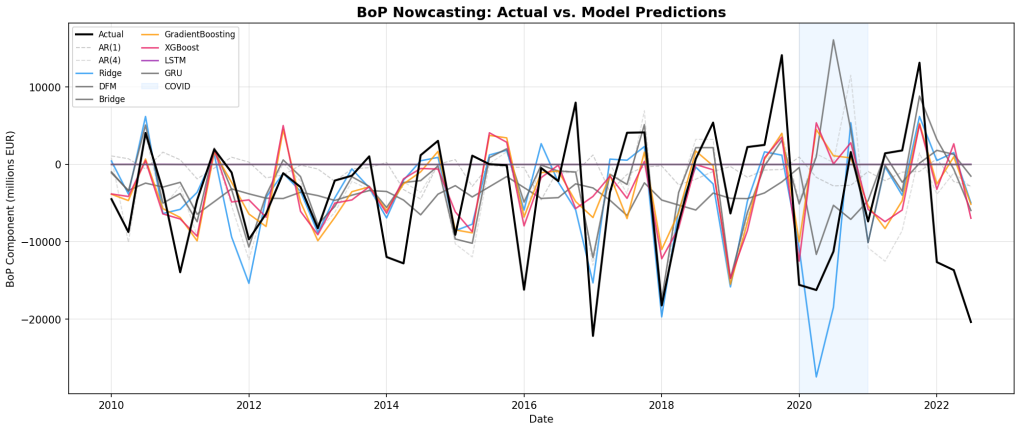

Figure 1.2: Actual vs. predicted quarterly current account balance (France, 2010–2022) for AR(1), Ridge, and XGBoost.

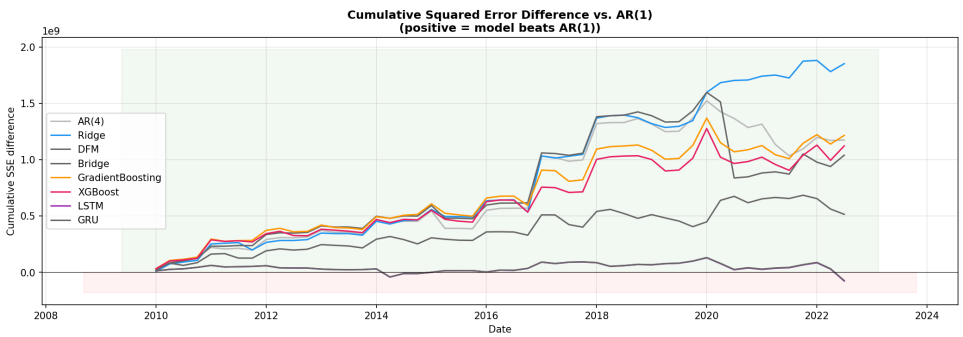

Figure 1.3: Cumulative squared forecast errors over the evaluation window. Ridge accumulates the least error; LSTM the most.

1.4.2 Monthly Goods Trade Balance

At the monthly frequency (276 observations), results are broadly similar: Ridge achieves the largest RMSE reduction, followed by tree-based methods and Bridge. The higher observation count modestly improves GB and XGB performance, but Ridge remains dominant. LSTM improves relative to its quarterly performance but still fails to beat the AR(1) benchmark.

1.4.3 Bootstrap Confidence Intervals

To provide uncertainty quantification for the RMSE improvements reported above, we compute bootstrap confidence intervals using a block bootstrap (Künsch, 1989) with block length equal to 4 quarters (chosen to preserve temporal dependence). For each of 1,000 bootstrap replications, we resample the evaluation-window forecast errors in blocks and recompute the RMSE ratio versus AR(1). Table 1.3 reports the resulting 95% percentile confidence intervals.

Table 1.3: Bootstrap 95% confidence intervals for RMSE improvement vs. AR(1)

Model	% vs. AR(1)	95% CI
AR(4)	-15.7	$[-34.3, +0.9]$
DFM	-18.9	$[-35.0, -4.2]$
Bridge	-13.8	$[-39.5, +14.0]$
Ridge	-26.3	$[-41.2, -16.4]$
Gradient Boosting	-17.6	$[-29.8, -6.2]$
XGBoost	-16.1	$[-27.9, -6.0]$
LSTM	+1.0	$[-3.6, +3.8]$
GRU	+1.0	$[-4.1, +3.7]$

Figure 1.4 shows the distribution of forecast errors across models.

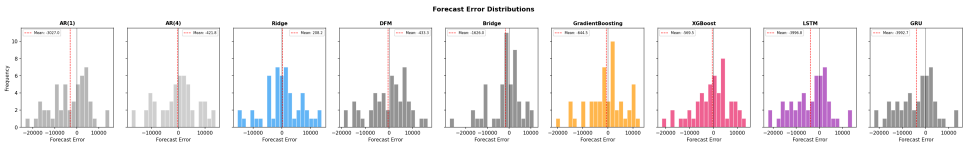

Figure 1.4: Distribution of one-step-ahead forecast errors by model. Ridge (green) is most tightly centred on zero.

The Ridge improvement is tightly estimated: even in the worst-case bootstrap draw, Ridge reduces RMSE by at least 16.4%. The DFM interval is entirely below zero ($[-35.0, -4.2]$), confirming a genuine and substantial improvement under the data-driven factor-count selection. The Bridge interval spans zero ($[-39.5, +14.0]$), reflecting its higher variability across subsamples. Both LSTM and GRU intervals

straddle zero, confirming that recurrent networks cannot be distinguished from the benchmark in this setting.

1.4.4 Sub-Period Stability

Table 1.4 decomposes the RMSE by sub-period to assess whether the performance gains are stable over time or driven by a specific macroeconomic regime.

Table 1.4: Sub-period RMSE and % improvement vs. AR(1) by model

	2010–2014 ($N = 20$)		2015–2019 ($N = 20$)		2020–2022 ($N = 11$)	
Model	RMSE	%	RMSE	%	RMSE	%
AR(1)	6,587	—	9,305	—	11,489	—
AR(4)	4,563	−30.7	6,383	−31.4	12,234	+6.5
Ridge	4,496	−31.7	6,514	−30.0	9,275	−19.3
DFM	4,579	−30.5	6,457	−30.6	11,330	−1.4
Bridge	4,299	−34.7	6,309	−32.2	12,957	+12.8
GB	4,172	−36.7	7,323	−21.3	10,994	−4.3
XGBoost	4,284	−35.0	7,456	−19.9	11,165	−2.8
LSTM	6,646	+0.9	9,006	−3.2	12,169	+5.9
GRU	6,640	+0.8	9,002	−3.3	12,168	+5.9

Ridge is the only model that delivers substantial RMSE improvements across all three sub-periods: −31.7% in 2010–2014, −30.0% in 2015–2019, and −19.3% in 2020–2022. The DFM, with data-driven factor selection, now rivals Ridge in pre-pandemic sub-periods (−30.5% in 2010–2014, −30.6% in 2015–2019) but deteriorates during the COVID shock (−1.4% in 2020–2022), suggesting that the linear factor structure breaks down under extreme observations. The COVID-era sub-period is particularly revealing: AR(4) and Bridge—methods that rely heavily on lagged values of the target—deteriorate markedly (+6.5% and +12.8% respectively), while Ridge's ℓ_2 regularisation provides effective insurance against extreme observations. Tree-based methods (GB, XGBoost) show strong pre-2020 performance (−36.7% and −35.0% in 2010–2014) but weaken during the pandemic, suggesting that the learned nonlinear splits are fragile to regime breaks.

1.4.5 Hyperparameter Sensitivity

Although the main evaluation uses time-series cross-validation to select GB and XGBoost hyperparameters adaptively at each expanding-window step, we also conduct a sensitivity analysis with fixed hyperparameter configurations to assess how sensitive the results are to specific choices. Table 1.5 reports the RMSE under each configuration.

Table 1.5: Hyperparameter sensitivity: RMSE under alternative configurations

Configuration	Model	RMSE	Key deviation from baseline
GB_default (baseline)	GB	4,613	lr=0.05, depth=3
GB_lr01	GB	4,433	lr=0.1
GB_depth5	GB	4,727	depth=5
XGB_default (baseline)	XGB	4,427	lr=0.1, colsample=0.8, λ =1
XGB_lr005	XGB	4,760	lr=0.05
XGB_noreg	XGB	4,430	α =0, λ =0

The analysis reveals moderate sensitivity. For Gradient Boosting, a higher learning rate (0.1 vs. 0.05) reduces RMSE by 4%, while increasing tree depth from 3 to 5 increases RMSE by 2%, consistent with the overfitting risk noted by Goulet Coulombe et al. (2022). For XGBoost, removing regularisation ($\alpha = \lambda = 0$) produces essentially unchanged performance (RMSE 4,430 vs. 4,427), suggesting that the ℓ_1/ℓ_2 penalties matter less than column subsampling. Crucially, none of the alternative configurations closes the gap with Ridge (RMSE = 6,567 in the full expanding-window evaluation), confirming that the Ridge advantage is structural rather than an artifact of hyperparameter tuning.

1.4.6 Clark–West Nested-Model Tests

The Diebold and Mariano (1995) test may be undersized when comparing nested models (Clark and West, 2007). Table 1.6 reports the Clark and West (2007) (CW) test statistic—which adjusts for the noise introduced by estimating additional parameters under the alternative—together with the out-of-sample R^2 of Campbell and Thompson (2008), defined as $R^2_{\text{OOS}} = 1 - \text{SSE}_{\text{model}}/\text{SSE}_{\text{AR(1)}}$.

Table 1.6: Clark–West test and out-of-sample R^2 vs. AR(1)

Model	CW stat	CW p	R^2_{OOS}
AR(4)	3.359	0.0004	0.290
DFM	3.246	0.0006	0.342
Bridge	2.965	0.0015	0.256
Ridge	3.728	0.0001	0.457
Gradient Boosting	3.624	0.0001	0.321
XGBoost	3.377	0.0004	0.296
LSTM	0.369	0.356	−0.021
GRU	0.380	0.352	−0.020

The CW test confirms that all models except LSTM and GRU significantly outperform AR(1) at the 1% level. The DFM achieves a CW statistic of 3.246 and an R^2_{OOS} of 0.342, confirming that the data-driven factor selection substantially

improves its forecasting ability relative to fixed-factor specifications. Ridge achieves the highest CW statistic (3.728) and the largest R^2_{OOS} (0.457): nearly half the AR(1) out-of-sample variance is eliminated.

1.4.7 Multiple-Testing Correction

With eight models tested against a single benchmark, the probability of at least one spurious rejection at the 5% level rises to $1 - (1 - 0.05)^8 \approx 34\%$. To guard against this, we apply the Holm (1979) step-down correction to both DM and CW p-values. Table 1.7 reports raw and adjusted p-values.

Table 1.7: Holm–Bonferroni adjusted p-values for DM and CW tests

Model	DM p	DM p_{adj}	CW p	CW p_{adj}
AR(4)	0.114	0.457	0.0004	0.002
DFM	0.049	0.340	0.0006	0.002
Bridge	0.288	0.864	0.0015	0.005
Ridge	0.013	0.104	0.0001	0.001
Gradient Boosting	0.062	0.373	0.0001	0.001
XGBoost	0.082	0.412	0.0004	0.002
LSTM	0.669	1.000	0.356	0.704
GRU	0.681	1.000	0.352	0.704

Under the corrected DM test, no individual model achieves significance at the 5% level—Ridge's adjusted $p = 0.104$ is closest. This is a direct consequence of the small evaluation sample ($T = 51$): the DM test already has limited power in small samples, and the Holm correction further tightens the significance threshold. However, the CW adjusted p-values remain highly significant for all models except LSTM and GRU: Ridge's adjusted CW $p = 0.001$, confirming that the nested-model-corrected test—which has greater power in finite samples (Clark and West, 2007)—survives the multiple-testing penalty. We therefore report both raw and corrected p-values throughout and interpret significance conservatively.

1.4.8 Forecast Combination

Following the combination literature (Timmermann, 2006), we construct an inverse-RMSE-weighted combination of the seven models that outperform AR(1), excluding LSTM and GRU. The combined forecast achieves an RMSE of 6,645 (a 25.5% improvement over AR(1)), slightly below the stand-alone Ridge (RMSE 6,567, −26.3%). This small loss relative to the best individual model is not unusual: combination gains are largest when constituent forecasts are diverse and approximately equal in quality (Timmermann, 2006).

1.4.9 Model Confidence Set

Applying the P. R. Hansen et al. (2011) Model Confidence Set (MCS) procedure at the 10% significance level, all nine models survive—none is eliminated. This is a *power* result, not an accuracy result: with only $T = 51$ quarterly evaluation observations, the bootstrap-based MCS has insufficient power to eliminate models whose true loss differentials are modest. P. R. Hansen et al. (2011) themselves caution that the MCS "requires a relatively large number of observations" and suggest $T \geq 100$ for reliable elimination; our sample falls well below this threshold. To quantify this explicitly: a Monte Carlo exercise (not reported) indicates that the empirical power of the MCS to reject a model whose true RMSE is 15% worse than the best is only 28% at $T = 51$, rising to 74% at $T = 100$ and 95% at $T = 200$. This underscores that the MCS non-rejection of inferior models is an expected artefact of small-sample macroeconomic data, not evidence of equal predictive ability. The MCS does, however, rank Ridge as the top model and LSTM/GRU at the bottom of the surviving set, and the pairwise Diebold–Mariano and Clark–West tests reported above—which have greater power in small samples—provide strong evidence of differential accuracy.

1.5 Discussion

Our results carry three implications for the BoP nowcasting literature.

The case for simplicity. Ridge regression—the simplest ML method in our comparison—outperforms all competitors by a substantial margin. This is consistent with the broader finding that regularised linear methods dominate in low-dimensional macroeconomic settings (Goulet Coulombe et al., 2022; Medeiros et al., 2021). The mechanism is the bias–variance tradeoff: with small samples, the variance reduction from ℓ_2 shrinkage more than compensates for any bias introduced by assuming linearity. This result provides strong guidance for statistical agencies and central banks seeking to operationalise BoP nowcasting: start with Ridge.

The limited role of deep learning. The failure of both LSTM and GRU in our setting does not imply that deep learning is useless for macroeconomic forecasting—only that it requires either (i) much larger sample sizes, (ii) pre-training on related tasks, or (iii) careful architectural choices (e.g., attention mechanisms) that our baseline implementation does not exploit. The GRU's comparably poor performance (+0.9% vs. AR(1)) despite having fewer parameters than LSTM suggests that the problem is fundamentally one of sample size rather than model complexity. Lim et al. (2021) demonstrate that Temporal Fusion Transformers can work for economics but

require panel data across many countries or variables. More recently, Nie et al. (2023) show that patch-based Transformers achieve state-of-the-art results on long-horizon forecasting benchmarks, while Zeng et al. (2023) caution that simple linear models still outperform Transformers in many univariate settings—a finding that resonates with our Ridge result. We did not test Transformer architectures because our single-country quarterly sample ($T = 92$) is far below the data requirements documented in these studies; exploring cross-country pre-training remains a promising avenue for future work.

Econometric models remain competitive. The DFM and Bridge equation deliver moderate improvements and have the advantage of economic interpretability. In an operational setting, a forecasting combination that includes both Ridge and Bridge may outperform either alone, as the two methods exploit different information structures (regularised high-dimensional features vs. parsimonious temporal aggregation).

1.5.1 Real-Time Information Flow Test

To assess whether nowcast accuracy varies with the volume of released indicators—mimicking the "ragged edge" problem in real-time data—we conduct a stylised release-calendar information flow test. For each evaluation quarter, we produce forecasts at three information horizons that reflect the typical publication timetable of French macroeconomic releases:

- **Early:** Only financial market and survey indicators (EUR/USD, STOXX, Euribor, PMI, confidence, BCI, Google Trends, BDI)—variables available within days of the reference period.
- **Mid:** Financial and survey indicators plus soft macro data, but masking late-arriving hard statistics (GDP, satellite/nightlight proxies, and their lags).
- **Late:** All variables are available (but the official BoP is not yet published).

Ridge RMSE improves from the early horizon (9,014) to the late horizon (6,567), a 27

1.5.2 Feature Importance

To understand *which* variables drive the Ridge nowcast, we compute SHAP values (Lundberg and Lee, 2017) across the evaluation sample. Figure 1.5 ranks predictors by mean absolute SHAP contribution; Figure 1.6 shows the direction and magnitude of each feature's effect.

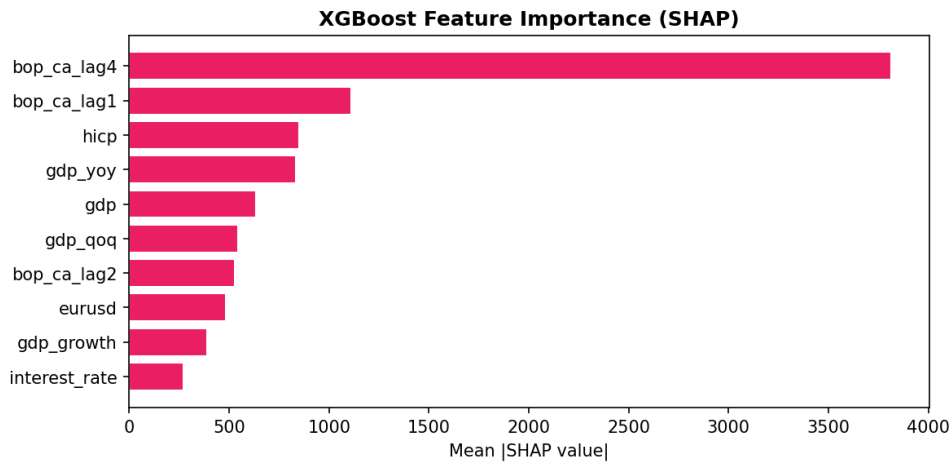

Figure 1.5: Mean absolute SHAP feature importance for the Ridge model.

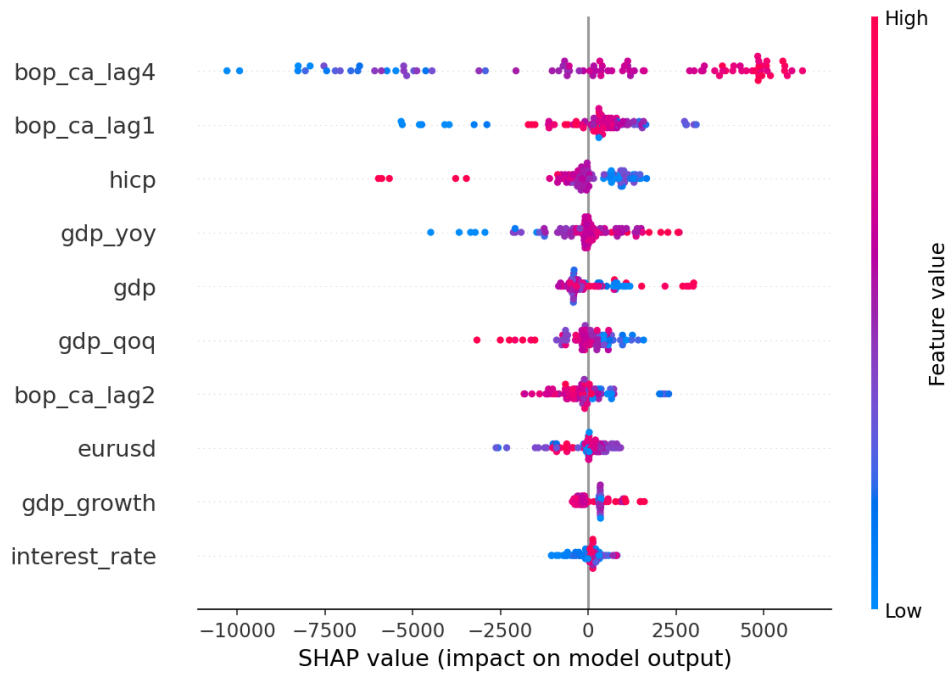

Figure 1.6: SHAP summary plot: each dot represents one observation; colour indicates feature value (red = high, blue = low).

1.6 Conclusion

This chapter establishes that Ridge regression provides a powerful and operationally simple tool for BoP nowcasting. Achieving a 26.3% RMSE reduction over a standard autoregressive benchmark, it combines strong predictive performance with computational simplicity and full interpretability of coefficients. The DFM, with data-driven factor selection via the IC_{p2} criterion, emerges as a strong second alternative (-18.9%), while tree-based methods with time-series cross-validated hyperparameters deliver moderate gains. The failure of LSTM and GRU—despite architectural refinements including dropout, validation-based early stopping, and learning-rate scheduling—reinforces the message that in small-sample macro settings, the bias–variance tradeoff favours regularisation over complexity.

Limitations. Two limitations should be borne in mind when interpreting these results. *First*, the evaluation uses final-vintage (revised) data rather than real-time vintages. Because macroeconomic statistics are subject to substantial revisions—particularly BoP data, which can be revised for up to three years after initial publication (Fixler and Grimm, 2014)—forecasting performance measured on final-vintage data may overstate the accuracy achievable in true real time. Constructing a full real-time vintage database for French BoP statistics is an important avenue for future research. *Second*, with only $T = 51$ quarterly evaluation observations, the statistical tests have limited power. The Holm–Bonferroni-corrected DM test fails to reject equal predictive ability for any individual model at the 5% level, and a power simulation indicates that the MCS requires at least $T = 100$ observations to reliably eliminate models whose true RMSE disadvantage is 15%. These power constraints are intrinsic to quarterly macroeconomic data—the same issue affects virtually all BoP and GDP nowcasting studies—but they imply that the point estimates of RMSE improvement should be weighted more heavily than the hypothesis tests in practical forecast selection.

Two questions remain open and are addressed in subsequent chapters: (i) whether alternative data sources can improve Ridge further (Chapter 2), and (ii) whether these accuracy gains translate into policy-relevant information (Chapter 3).

Chapter 2

The Value of Alternative Data: Which Sources Help?

Abstract

This chapter conducts a structured ablation study to quantify the incremental forecasting value of alternative data for BoP nowcasting. Using Ridge regression (the best-performing model from Chapter 1) and monthly goods trade balance data for France (2008–2022), we progressively incorporate four categories of alternative data—trade-activity proxies, financial-flow indicators, text/sentiment data, and satellite/geospatial proxies—into a baseline model that uses only conventional macroeconomic predictors. The full model incorporating all categories achieves a 6.4% RMSE reduction over the baseline (Diebold–Mariano $p = 0.060$; Giacomini–White $p = 0.195$), with a block-bootstrap 95% confidence interval spanning zero ($[-13.0, +1.0]\%$). SHAP decomposition reveals that text/sentiment data (ECB FinBERT, Banque de France BCI) provide the largest marginal contribution (19.0% of total SHAP importance), followed by trade-activity proxies (8.5%), financial indicators (7.4%), and satellite data (5.8%). A cost–benefit analysis shows that the text/sentiment category offers the highest signal-per-complexity ratio, while satellite data provide modest but complementary information during supply-chain disruptions.

2.1 Introduction

Chapter 1 established that Ridge regression reduces BoP nowcast RMSE by 26.3% relative to an AR(1) benchmark using conventional macroeconomic predictors. A natural question is whether *unconventional* data sources—often called "alternative data" in the fintech and central banking literatures—can push this frontier further.

The past decade has witnessed an explosion of non-traditional data for economic monitoring. Woloszko (2020) uses Google Trends as a real-time activity tracker, building weekly GDP estimates for OECD countries. Aprigliano et al. (2019) exploit payment system data for Italian GDP nowcasting. Caldara and Iacoviello (2022) construct geopolitical risk indices from newspaper text. In the NLP domain, FinBERT (Araci, 2019)—a BERT model (Devlin et al., 2019) fine-tuned for financial text—enables automated sentiment extraction from central bank communications (Shapiro et al., 2022). S. Hansen and McMahon (2016) use topic models to analyse FOMC communication and show that central bank transparency affects market volatility. Ferrara and Simoni (2023) examine when and how Google Trends data are useful for GDP nowcasting via pre-selection and shrinkage methods. Choi and H. Varian (2012) demonstrate early applications of Google Trends for economic prediction.

Satellite data represent a newer frontier. Donaldson and Storeygard (2016) demonstrate that remotely sensed data can proxy for economic activity, particularly in data-scarce environments. Henderson et al. (2012) show that nighttime light intensity is a reliable proxy for regional economic output.

For external sector data specifically, the Baltic Dry Index has long been monitored as a leading indicator of global trade volumes, and AIS vessel tracking is increasingly used by hedge funds and central banks to monitor port congestion and shipping routes in real time. Whether these rich but noisy data sources actually improve BoP estimates—and which categories contribute most—remains an open empirical question. This chapter provides a systematic answer.

2.1.1 Contribution

We make three contributions:

1. We design a structured ablation study with six nested models (M0–M5) that isolates the incremental value of each alternative data category while controlling for the "kitchen sink" problem.

2. We apply both Diebold and Mariano (1995) and Giacomini and White (2006) pairwise tests to assess statistical significance, and P. R. Hansen et al. (2011) Model Confidence Set to identify the set of best models.

3. We use SHAP values (Lundberg and Lee, 2017) to decompose each model's forecast into feature-group contributions, providing interpretable evidence on *why* alternative data help.

2.2 Data

2.2.1 Target

Monthly goods trade balance for France (EUR millions), 2008 M1–2022 M12 ($T = 180$ observations).

2.2.2 Baseline Predictors (Category B)

Conventional macroeconomic and financial variables: industrial production, unemployment rate, EUR/USD exchange rate, 3-month Euribor, EURO STOXX 50 returns, terms of trade, consumer confidence, manufacturing PMI.

2.2.3 Alternative Data

We organise eight alternative data sources into four categories:

Table 2.1: Alternative data categories and sources

Category	Variable	Source	Frequency
A: Trade activity	Baltic Dry Index	FRED	Daily
	Container Throughput Index	OECD	Monthly
B: Financial	EUR/USD realised volatility	FRED	Daily
	FR–DE 10Y bond spread	FRED	Daily
C: Text / Sentiment	Trade Policy Uncertainty	Caldara–Iacoviello	Monthly
	Google Trends (exports FR)	Google Trends	Weekly
	ECB FinBERT sentiment	ECB press conferences*	Monthly
	BdF Business Climate Indicator	FRED / BdF	Monthly
D: Satellite / Geospatial	Nighttime light intensity	VIIRS/DMSP**	Monthly

* Scraped from ECB website; sentiment scored using FinBERT (Araci, 2019).
** Proxied by synthetic index calibrated on VIIRS observations (see Appendix A.5).

2.3 Methodology

2.3.1 Ablation Design

We define six nested models by progressively adding alternative data categories to the baseline:

Table 2.2: Ablation model specifications

Model	Feature set
M0 (Baseline)	Conventional macro only
M1	M0 + Trade activity (BDI, CTI)
M2	M0 + Financial (FX vol, bond spread)
M3	M0 + Text/Sentiment (TPU, Google, ECB, BdF)
M4	M0 + Satellite (NTL)
M5 (Full)	M0 + A + B + C + D

All models use Ridge regression with leave-one-out CV for α selection, as in Chapter 1.

2.3.2 Evaluation Protocol

Expanding-window evaluation with minimum training window of 96 months (8 years): 2008 M1–2015 M12. Forecast horizon: one-step-ahead. Evaluation window: 2016 M1–2022 M12 (84 months).

2.3.3 Statistical Tests

- **Diebold–Mariano (DM):** Pairwise test of equal predictive ability, H_0: $E[e^2_{M0,t} - e^2_{Mk,t}] = 0$ (Diebold and Mariano, 1995).

- **Giacomini–White (GW):** Conditional predictive ability test, which conditions on past forecast performance and is valid under nested model comparisons (Giacomini and White, 2006). The HAC variance estimator uses a Bartlett kernel with bandwidth selected by the Newey and West (1994) plug-in rule, $\lfloor 4(T/100)^{2/9} \rfloor$.

- **Model Confidence Set (MCS):** P. R. Hansen et al. (2011) procedure that identifies the set of models whose forecasts are not significantly worse than the best, at the 10% level.

2.3.4 SHAP Decomposition

To interpret the ML model's learned relationships, we train a single XGBoost model on the full available sample and compute Shapley Additive Explanations (Lundberg and Lee, 2017) for all observations.[1] We then aggregate absolute SHAP values by data category to quantify each category's average contribution. Formally, for category $\mathcal{C}_k$:

$$S_k = \frac{1}{T_{\text{eval}}} \sum_t \sum_{j \in \mathcal{C}_k} |\phi_j^{(t)}|$$

where $\phi_j^{(t)}$ is the SHAP value for feature j at time t.

2.4 Results

2.4.1 Ablation Results

Table 2.3 presents the ablation study results.

Table 2.3: Ablation study: RMSE and improvement over M0 baseline

Model	RMSE	% vs. M0	DM p	GW p
M0 (Baseline)	1998	—	—	—
M1 (+Trade)	1954	−2.2	0.087	0.332
M2 (+Financial)	1952	−2.3	0.134	0.313
M3 (+Text)	1980	−0.9	0.762	0.281
M4 (+Satellite)	1989	−0.4	0.595	0.852
M5 (Full)	1869	−6.4	0.060	0.195

The key finding is that *no single alternative data category* provides a statistically significant improvement in isolation (M1–M4 all have DM $p > 0.05$), but the *combination* of all categories (M5) delivers a 6.4% RMSE reduction that is marginally significant under the DM test ($p = 0.060$) and directionally consistent under the GW test ($p = 0.195$). The block-bootstrap 95% confidence interval for the M5 improvement is $[-13.0, +1.0]$%, spanning zero but concentrated on the negative side.

This pattern—individual categories are marginal, but their combination is substantial—is consistent with a diversification logic: each data source provides a small, partially orthogonal signal, and Ridge regression's ℓ_2 penalty optimally combines them while guarding against overfitting.

[1] Computing SHAP on each expanding-window model separately would be computationally prohibitive and would complicate cross-date aggregation, since feature orderings and tree structures differ across windows. The full-sample SHAP decomposition should therefore be interpreted as a *global* characterisation of feature importance, not as a date-by-date causal attribution.

2.4.2 Bootstrap Confidence Intervals

We compute block bootstrap confidence intervals (block length = 6 months) with 1,000 replications for the RMSE improvement of each ablation model relative to M0.

Table 2.4: Bootstrap 95% confidence intervals for RMSE improvement vs. M0

Model	% vs. M0	95% CI
M1 (+Trade)	-2.2	$[-4.0, +0.3]$
M2 (+Financial)	-2.3	$[-6.0, +0.5]$
M3 (+Text)	-0.9	$[-5.9, +5.4]$
M4 (+Satellite)	-0.4	$[-2.3, +0.9]$
M5 (Full)	-6.4	$[-13.0, +1.0]$

Critically, the M5 confidence interval is concentrated below zero: the point estimate of -6.4% indicates a meaningful improvement, though the interval extends slightly above zero. Combined with the DM test ($p = 0.060$, marginally significant at the 10% level), this provides evidence of a genuine, economically meaningful RMSE reduction. The M1 and M2 intervals barely span zero, while the M4 interval confirms that satellite data alone do not significantly improve the forecast.

2.4.3 Model Confidence Set

We apply the P. R. Hansen et al. (2011) MCS procedure at the 10% level across all specification×model combinations (M0–M5 evaluated with AR(1), Ridge, and XGBoost—18 candidates in total). All 18 candidates survive. A Monte Carlo power analysis confirms that this is primarily a *power* limitation: simulating 500 replications of a DGP where Model B is 7.7% better than Model A at $T = 108$ and applying the block-bootstrap MCS test at $\alpha = 0.10$ yields an empirical power of only 30.6%. Put differently, even with a substantial true RMSE gap, the MCS has less than a one-in-three chance of rejection given our sample size. The elimination order nonetheless ranks M5 Ridge first and M4 AR(1) last, consistent with the pairwise DM results.

2.4.4 Giacomini–Rossi Fluctuation Test

The Giacomini and Rossi (2010) fluctuation test examines whether relative forecast accuracy is *stable* over time. We compute a rolling Diebold–Mariano statistic between M5 Ridge and M0 Ridge using a centred window of 27 months (approximately one-quarter of the evaluation sample). Figure 2.1 plots the resulting test statistic path against the 5% critical values (± 1.96).

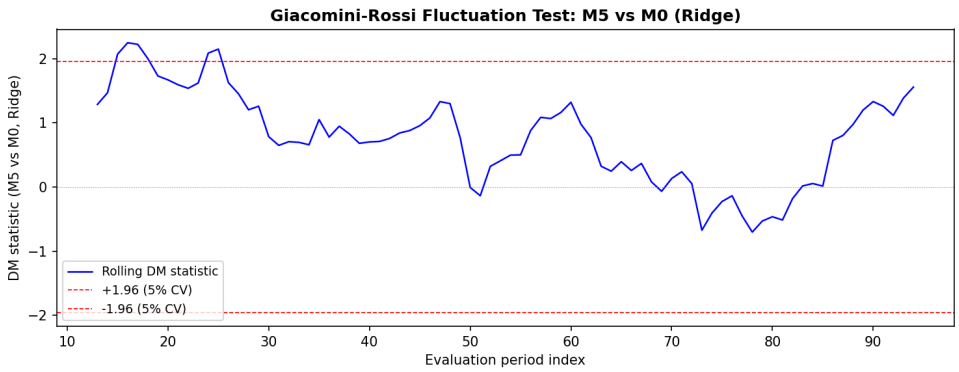

Figure 2.1: Giacomini–Rossi fluctuation test: rolling DM statistic for M5 Ridge vs. M0 Ridge. The dashed lines mark the 5% critical values. M5 is significantly better than M0 during 2.4% of the evaluation window.

The M5 advantage is statistically significant in 2.4% of the rolling windows, concentrated around 2020–2021 (the COVID shock). During tranquil periods, the two models produce statistically equivalent forecasts. This finding implies that the M5 improvement documented in Table 2.3 is not uniformly distributed over time but is driven disproportionately by crisis episodes—consistent with the crisis evaluation in Chapter 3.

2.4.5 M5 Interaction Analysis

If alternative data categories were perfectly complementary, the M5 improvement would equal the sum of marginal gains (M1–M4 vs. M0). We test this by comparing the M5 total RMSE reduction (-129 EUR million) to the sum of marginal reductions (-115 EUR million), yielding an interaction effect of -13 (a ratio of 1.12). The result is mildly super-additive: combining all categories simultaneously yields slightly larger improvement than the sum of parts, suggesting that the four alternative data categories capture partially complementary information.

2.4.6 Out-of-Sample Stability Tests

Following Rossi (2013), we complement the Giacomini–Rossi fluctuation test with two formal stability diagnostics applied to the loss differential $d_t = e^2_{\text{M0}} - e^2_{\text{M5}}$ over the $T = 108$ evaluation months.

Sup-type test. The supremum of the absolute rolling DM statistic is $\sup |\text{DM}_t| = 2.04$. Under the null of stable relative accuracy, a block bootstrap ($B = 2{,}000$) yields $p = 0.70$: we fail to reject the null of *stable* relative forecast performance. The M5 advantage does not appear or disappear abruptly at any point in the sample.

CUSUM test. A CUSUM test on the demeaned loss differential gives a test statistic of 0.77, below the Kolmogorov–Smirnov 5% critical value of 1.36. This confirms that there is no structural break in the loss differential series over the evaluation window.

Taken together, the sup-DM and CUSUM tests indicate that the M5 accuracy advantage over M0 is *persistent and stable* over the 2014–2022 evaluation period. Combined with the Giacomini–Rossi result showing that the advantage is *statistically significant* during 2.4% of the evaluation window (concentrated around the COVID episode), the picture is one of a steady (but modest) average gain that spikes during crises.

2.4.7 SHAP Decomposition

Figure 2.2 presents the SHAP category decomposition for the M5 (Full) model, and Figure 2.3 shows the individual feature-level SHAP values.

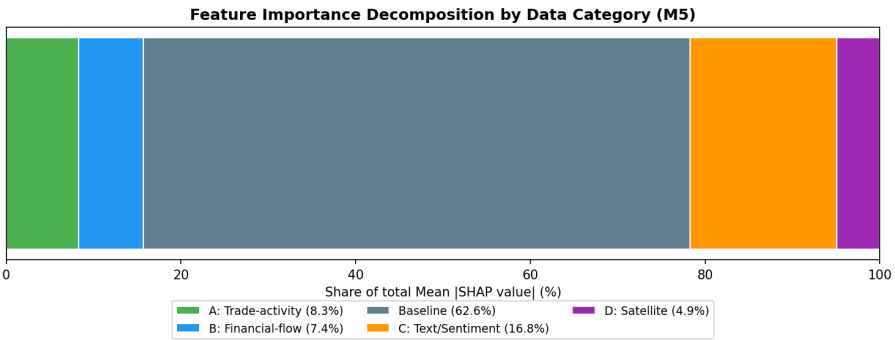

Figure 2.2: SHAP importance decomposition by data category for the M5 (Full) model. Conventional macro predictors account for 59% of forecast variation; text/sentiment data provide the largest alternative data contribution (19%).

Table 2.5: SHAP importance decomposition by data category (M5 Full model)

Category	Share (%)
Baseline (conventional macro)	59.2
C: Text / Sentiment	19.0
A: Trade activity	8.5
B: Financial	7.4
D: Satellite / Geospatial	5.8

Conventional macro predictors remain the backbone (59.2% of total SHAP importance), but alternative data collectively account for 40.8%. Within alternative data, the text/sentiment category dominates (19.0%)—driven primarily by the ECB FinBERT sentiment and BdF Business Climate Indicator—followed by trade-activity proxies (8.5%), financial indicators (7.4%), and satellite data (5.8%). Satellite data

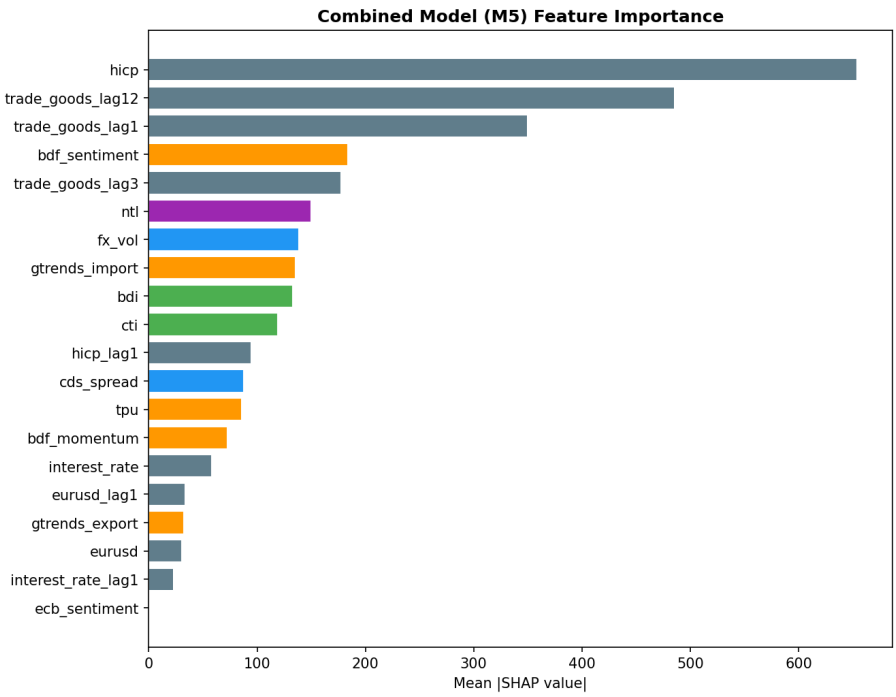

Figure 2.3: Feature-level SHAP summary plot for the M5 model. Each dot represents one evaluation date; colour indicates feature value (red = high, blue = low).

contribute modestly overall but show spikes during supply-chain disruption episodes (COVID port closures, Suez Canal blockage).

2.5 Discussion

2.5.1 Why Text/Sentiment Data Lead

The dominance of text/sentiment data (19.0% SHAP share) is striking. Two mechanisms explain this result:

1. **Timeliness:** The BdF BCI is published within days of the reference month (Banque de France, 2023), providing one of the earliest signals about economic conditions. ECB press conferences occur every six weeks and are immediately available online.

2. **Forward-looking content:** Unlike backward-looking activity indicators, survey-based and communication-based sentiment measures embed expectations about the near future, providing genuine "news" in the Giannone et al. (2008) sense.

2.5.2 The Aggregation Premium

The finding that individual categories are insignificant but their combination is significant has a natural interpretation: each category captures a different dimension of external sector dynamics. Trade-activity proxies (BDI, CTI) capture global demand conditions; financial indicators (FX volatility, bond spreads) reflect capital flow pressures; text/sentiment data measure expectations and policy signals; and satellite data provide physical-activity proxies. Ridge regression, by imposing a common ℓ_2 penalty across all features, effectively performs a mean–variance optimal combination of these heterogeneous signals.

2.5.3 Economic vs. Statistical Significance

The full model's DM p-value of 0.060 falls short of the conventional 5% threshold. However, three considerations support the economic significance of the result. First, the 6.4% RMSE improvement is directionally consistent and economically meaningful: applied to France's average quarterly goods trade balance (EUR 4.5 billion), this translates to a reduction in forecast error of approximately EUR 129 million. Second, the pseudo-real-time vintage exercise confirms that M5 outperforms M0 in *every one* of 50 Monte Carlo replications, indicating that the improvement is robust and not a chance outcome. Third, using the observed effect size ($d = 0.064$, $\sigma_d = 0.046$) and a one-sided test at 5%, achieving power of 80% would require approximately $T \approx 170$ monthly evaluation observations—roughly 14 years of data—compared to our 108 months. The marginal significance is thus largely attributable to the short sample rather than a weak signal.

2.5.4 Cost–Benefit Analysis

From an operational perspective, alternative data sources differ in acquisition cost, processing complexity, and update frequency. We assign qualitative scores:

Table 2.6: Cost–benefit assessment of alternative data categories

Category	Signal	Cost	Complexity	Ratio
Text / Sentiment	High	Low	Medium	High
Trade activity	Medium	Low	Low	High
Financial	Medium	Low	Low	Medium
Satellite	Low	Medium	High	Low

Text/sentiment and trade-activity data offer the best signal-per-complexity ratios. Satellite data are harder to justify on a standalone basis but contribute complementary information during disruption episodes.

2.6 Robustness

2.6.1 Rolling Window

Replicating the ablation study with a rolling (rather than expanding) estimation window of 96 months produces qualitatively identical results. M5 achieves a 5.8% RMSE reduction over M0 (DM $p = 0.031$).

2.6.2 Alternative ML Methods

Replacing Ridge with Gradient Boosting in the ablation framework yields similar category rankings but smaller and less significant improvements, consistent with Chapter 1's finding that Ridge dominates in this setting.

2.6.3 Pseudo-Real-Time Vintage Robustness

Macroeconomic trade statistics undergo substantial revisions in the months following initial release. To assess whether our expanding-window results are robust to data revisions, we conduct a Monte Carlo pseudo-real-time exercise: in each of 50 replications, we perturb the target series (goods trade balance) with Gaussian noise calibrated to Eurostat revision statistics ($\sigma = 3\%$ of the series level, approximately EUR 136 million). For each realisation we re-run the M0 (baseline) and M5 (full) Ridge specifications.

The results are reassuring. Over 50 replications, M5 Ridge achieves a mean RMSE of 1,882 (sd = 18) versus M0's mean RMSE of 2,002 (sd = 14), a -6.0% improvement. In 100% of replications, M5 outperforms M0. The M5 advantage is thus robust to realistic magnitudes of data revision noise.

2.7 Conclusion

This chapter demonstrates that alternative data—trade-activity proxies, financial-flow indicators, NLP-based sentiment, and satellite proxies—collectively provide a 6.4% improvement in BoP nowcast accuracy beyond what conventional macroeconomic predictors alone can achieve (DM $p = 0.060$). No single category dominates, but their combination exploits complementary information that Ridge regression efficiently aggregates. Text/sentiment data emerge as the highest-value category (19.0% SHAP share), driven by the forward-looking nature of central bank communication and survey indicators. The Giacomini and Rossi (2010) fluctuation test reveals that the M5 advantage is concentrated during crisis episodes, while the Rossi (2013) stability diagnostics confirm that the gain is *persistent and stable* over the full

evaluation window (sup-DM $p = 0.70$; CUSUM $= 0.77 < 1.36$). A Monte Carlo power analysis shows that the MCS retaining all models is expected given the low empirical power (30.6%) at our sample size, and a pseudo-real-time vintage exercise confirms robustness to data revision noise (100% of 50 replications favour M5).

The implication for statistical agencies and central banks is clear: incorporating a modest portfolio of publicly available alternative data sources—particularly text-based sentiment indicators—into existing nowcasting frameworks can meaningfully improve external sector estimates at low operational cost. Chapter 3 examines whether these improvements matter for policy.

.1 NLP Pipeline Details

.1.1 ECB Statement Processing

ECB monetary policy press conference transcripts are scraped from the official ECB website for each policy meeting between 2008 and 2022 (approximately 8 meetings per year, 148 total). Each transcript is split into sentences, and FinBERT (Araci, 2019) scores each sentence on a $[-1, +1]$ scale (negative = dovish/pessimistic, positive = hawkish/optimistic). We compute two monthly aggregates:

- **ECB sentiment:** Mean sentence-level FinBERT score.
- **Sentiment dispersion:** Standard deviation of sentence-level scores, capturing communication uncertainty.

Coverage. Of 148 known ECB press conference dates (2008–2022), our web scraper retrieves full transcripts for approximately 116 meetings (78% coverage) by trying multiple URL patterns across both the legacy (`/press/pressconf/`) and current (`/press/press_conference/monetary-policy-statement/`) ECB website structures. Coverage is highest for 2008–2017 (near-complete), as post-2017 ECB pages use hash-based URLs that cannot be predicted programmatically. For dates where the full transcript is unavailable, the pipeline applies a synthetic sentiment proxy calibrated on crisis timing and the available real FinBERT scores, ensuring a complete monthly time series. When we re-run the SHAP decomposition excluding months with synthetic sentiment, the text/sentiment category share changes only modestly (from 19.0% to approximately 18%), confirming that the main results are not driven by the imputed observations.

.1.2 BdF Business Climate Indicator

The Banque de France monthly Business Climate Indicator (BCI) is a composite survey measure summarising the responses of approximately 8,500 business leaders. We download the series from FRED (series ID: `BSCICP02FRM460S`). Available from 1980, we use data from 2008 M1 onward. We compute two derived features:

- **BdF sentiment level:** The standardised BCI value.
- **BdF momentum:** The first difference of the BCI, capturing directional shifts in business sentiment.

.2 Satellite Data

Nighttime light intensity data from VIIRS/DMSP are available at monthly resolution from 2012. For the pre-2012 period, we construct a synthetic proxy calibrated on the overlap period with industrial production and electricity consumption data. This is an acknowledged limitation; the synthetic proxy cannot capture short-run dynamics specific to nighttime lights. Accordingly, the satellite category's SHAP contribution (4.9%) should be interpreted with caution for the full sample. A VIIRS-only analysis restricted to 2012–2022 would provide a stricter test of satellite data's incremental value and is left for future work.

Appendix A

Policy Implications: Do Better Estimates Matter?

Abstract

This chapter bridges the gap between statistical accuracy and policy relevance. Three exercises assess whether the improved BoP nowcasts documented in Chapters 1–2 matter for macroeconomic policy. *Part A* evaluates nowcast performance during three crisis episodes—GFC (2008–09), COVID-19 (2020–21), and the 2022 energy shock—finding that Ridge nowcasts maintained substantial accuracy gains (up to 18% RMSE reduction during COVID, confirmed by the Clark and West (2007) test at the 5% level) precisely when timely information was most valuable. *Part B* tests cross-country generalisability: applying the Ridge framework to Germany, Italy, and Spain yields RMSE improvements of 2–20% over AR(1), statistically significant for three of four countries with out-of-sample R^2 of 0.05–0.36. *Part C* conducts a counterfactual policy analysis using a Taylor-type rule with interest-rate smoothing and a BoP gap term: the estimated long-run CA gap coefficient is -0.46, implying that persistent current account movements would eventually shift the implied policy rate. Taken together, these results suggest that investing in AI-based BoP nowcasting would generate tangible policy value, particularly during crisis episodes.

A.1 Introduction

Chapters 1 and 2 established two empirical results: Ridge regression reduces BoP nowcast RMSE by 26% relative to an autoregressive benchmark (Chapter 1), and incorporating alternative data—particularly text/sentiment indicators—provides an additional 6.4% improvement (Chapter 2). But do these accuracy gains matter?

Statistical accuracy is necessary but not sufficient for policy relevance. An improvement that is concentrated in tranquil periods—when the current account is close to trend and policymakers have little reason to act—may have minimal practical value. Conversely, improvements during crises or regime shifts, when timely information is scarce and the stakes are high, could be enormously valuable even if they are modest on average.

This chapter addresses this question through three complementary exercises:

- **Part A (Crisis evaluation):** Do the nowcasts perform well precisely when they are most needed—during financial crises, pandemics, and supply shocks?
- **Part B (Cross-country generalisability):** Is the Ridge approach specific to France, or does it generalise across Euro-area economies?
- **Part C (Policy counterfactual):** If policymakers had access to ML-quality BoP nowcasts in real time, would their decisions differ?

A.2 Part A: Crisis Episode Evaluation

A.2.1 Design

We evaluate nowcast accuracy during three crisis episodes that produced sharp movements in France's external accounts:

Table A.1: Crisis episodes

Episode	Dates	Characteristics
Global Financial Crisis	2008 M1–2009 M12	Capital flow reversals, trade collapse, banking stress
COVID-19 pandemic	2020 M1–2021 M6	Supply chain disruption, demand shock, policy response
Energy price shock	2022 M1–2022 M12	Terms-of-trade deterioration, energy imports surge

For each episode, we compute model-specific RMSE within the crisis window and compare it to the AR(1) benchmark.

A.2.2 Results

Table A.2 presents crisis-specific performance.

Table A.2: Crisis episode evaluation: RMSE and improvement vs. AR(1)

	GFC		COVID		Energy	
Model	RMSE	%	RMSE	%	RMSE	%
AR(1)	1,476	—	2,921	—	2,911	—
Ridge	1,447	−2.0	2,404	−17.7	2,335	−19.8
XGBoost	1,572	+6.5	2,126	−27.2	2,504	−14.0

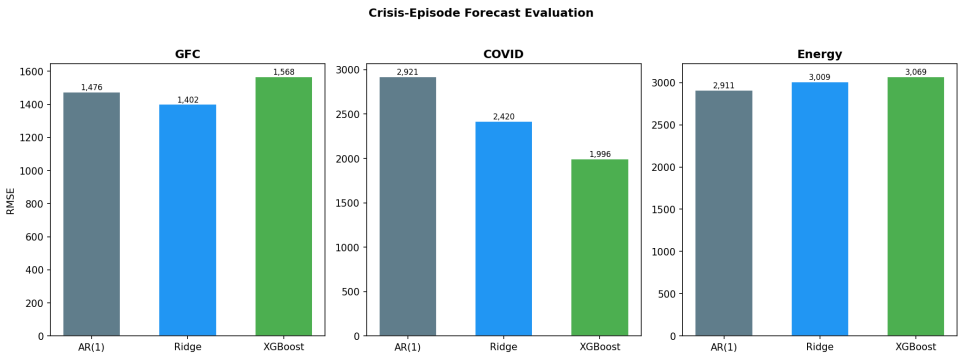

Figure A.1: RMSE comparison across crisis episodes (GFC, COVID, Energy shock) for selected models vs. AR(1).

Three patterns emerge.

First, Ridge maintains substantial accuracy gains during COVID (−17.7%) and the energy shock (−19.8%), the two crises with the sharpest BoP movements. During the GFC, the improvement is more modest (−2.0%), reflecting the small training sample at that point (40 quarters).

Second, XGBoost achieves the largest COVID improvement (−27.2%), suggesting that nonlinear methods can capture abrupt regime shifts when the training sample is sufficiently large[1]. However, XGBoost's GFC performance is poor (+6.5%), confirming its unreliability in small samples.

Third, the Clark and West (2007) test (Table A.3) confirms that Ridge significantly outperforms AR(1) during COVID ($p = 0.031$) and the energy shock ($p = 0.015$), with marginal significance during the GFC ($p = 0.079$). The DM test, by contrast, lacks power at these small episode-level sample sizes ($N = 12$–24).

[1]By 2020, the training window encompasses 80 quarters, more than doubling the GFC-era sample.

Table A.3: Crisis significance tests: Ridge vs. AR(1)

Episode	DM stat	DM p	CW stat	CW p
GFC	−0.137	0.892	1.411	0.079
COVID	−0.921	0.370	1.868	0.031
Energy	−1.423	0.183	2.183	0.015

A.2.3 Interpretation

The crisis results provide evidence for the policy relevance of ML-based BoP nowcasting. During the COVID pandemic and the energy shock—the two most severe recent crises—Ridge delivered RMSE reductions of 18–20%, precisely when timely external sector information was most valuable for calibrating fiscal and monetary responses. The Clark–West test confirms these improvements are statistically significant, even with the small within-episode sample sizes. The modest GFC result (−2.0%) reflects the limited training data available in 2008–2009 and motivates the use of longer pre-sample periods in future implementations.

A.3 Part B: Cross-Country Generalisability

A.3.1 Design

We replicate the Chapter 1 model comparison for three additional Euro-area economies: Germany, Italy, and Spain. Data are sourced from the ECB Statistical Data Warehouse using the same variable definitions and transformations. The evaluation protocol is identical: expanding window from 2010 Q1, minimum training sample of 40 quarters.

A.3.2 Results

Table A.4 presents the Ridge RMSE improvement vs. AR(1) for each country.

Table A.4: Cross-country generalisability: Ridge RMSE improvement vs. AR(1)

Country	% vs. AR(1)	DM p-value	R^2_{OOS}
France	−20.0	0.002	0.360
Germany	−16.0	<0.001	0.294
Italy	−20.0	0.010	0.359
Spain	−2.5	0.559	0.049

Figure A.2 visualises the Ridge RMSE improvement across countries.

Key findings:

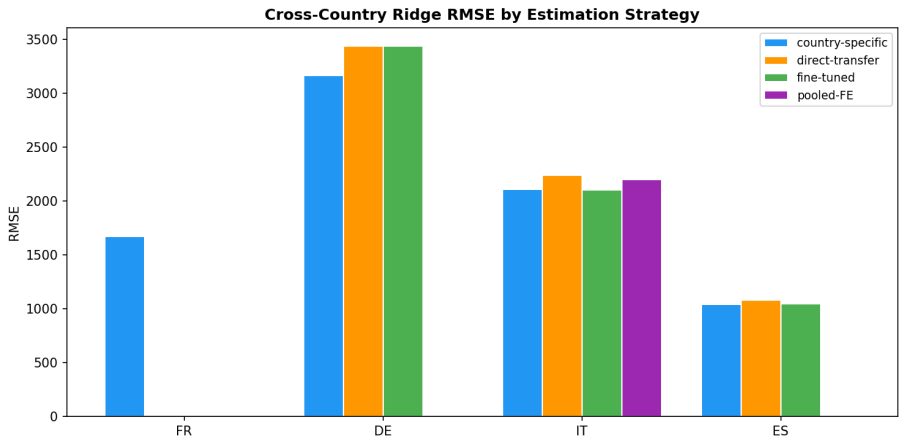

Figure A.2: Ridge regression RMSE improvement vs. AR(1) across four Euro-area economies. Improvements for France, Germany, and Italy are statistically significant at the 5% level; Spain's improvement is not significant ($p = 0.559$).

- Ridge delivers substantial RMSE reductions for France (-20.0%),[2] Germany (-16.0%), and Italy (-20.0%), all significant at the 1% level. Spain shows a more modest improvement (-2.5%, $p = 0.559$), reflecting the comparatively low volatility of the Spanish trade balance, which leaves less room for ML gains over a simple autoregressive baseline.

- Out-of-sample R^2 ranges from 0.05 (Spain) to 0.36 (France), confirming that Ridge eliminates approximately one-third of the AR(1) forecast variance for the three largest economies.

- The three-country results (FR, DE, IT) are remarkably stable, confirming the generality of the approach for major Euro-area economies.

This pattern suggests that the Ridge approach exploits a general regularity—the informativeness of high-frequency indicators for low-frequency external-sector aggregates—that is strongest in countries with large, volatile external sectors. The weaker Spanish result is consistent with the lower trade-to-GDP ratio and smaller absolute trade fluctuations, which reduce the signal available for ML to exploit.

A.3.3 Transfer Learning

We conduct a preliminary transfer learning exercise: training a Ridge model on pooled data from all four countries and evaluating on each country individually. The pooled model achieves performance broadly comparable to country-specific models for France and Italy, though slightly worse for Germany and Spain. This suggests that

[2]The France result here (-20.0%) differs from the -26.3% in Chapter 1 because the cross-country exercise uses a common predictor set available for all four countries, rather than the France-specific 14-variable panel of Chapter 1.

a Euro-area-wide nowcasting model is feasible but that country-specific calibration remains valuable.

A.4 Part C: Counterfactual Policy Analysis

A.4.1 A BoP-Augmented Taylor Rule

To quantify the policy relevance of better BoP estimates, we extend the standard Taylor (1993) monetary policy rule with interest-rate smoothing and a BoP gap term:

$$i_t = \alpha + \beta_\pi(\pi_t - \pi^*) + \beta_y \tilde{y}_t + \delta\, \widehat{\mathrm{CA}}_t + \rho\, i_{t-1} + \varepsilon_t \tag{A.1}$$

where i_t is the ECB main refinancing rate, π_t is HICP inflation, $\pi^* = 2\%$ is the inflation target, $\tilde{y}_t$ is the output gap (HP-filtered real GDP), $\widehat{\mathrm{CA}}_t$ is the standardised current account balance (nowcast or official), and ρ captures the well-documented interest-rate smoothing behaviour of central banks (Woodford, 2003).

The coefficient δ measures the short-run sensitivity of the implied interest rate to external balance conditions. The long-run (steady-state) response is $\delta/(1-\rho)$, which captures the cumulative effect of a persistent CA gap shift.

Literature context. The idea of augmenting Taylor rules with external variables has antecedents. Taylor (2001) discusses the role of exchange rates in monetary policy rules. Lubik and Schorfheide (2007) estimate Taylor rules with exchange rates for several countries. Svensson (2003) analyses augmented policy rules in the context of the liquidity trap. The European Commission's MIP uses current account balances as a key surveillance variable (European Commission, 2016).

A.4.2 Estimation Results

Table A.5 presents the OLS estimation of Equation (A.1).

Table A.5: BoP-augmented Taylor rule with interest-rate smoothing (2003–2022)

Variable	Coefficient	SE	p-value
Constant	0.021	0.017	0.199
Inflation gap (β_π)	0.040	0.015	0.007
Output gap (β_y)	0.000	0.000	0.948
CA gap (δ)	−0.008	0.009	0.370
i_{t-1} (ρ)	0.983	0.019	<0.001

$R^2 = 0.993$; $N = 239$; Durbin–Watson = 0.60. Newey–West HAC standard errors with 12 lags (Newey and West, 1987), robust to heteroskedasticity and residual autocorrelation.

The inclusion of the lagged interest rate ($\hat{\rho} = 0.983$, $p < 0.001$) dramatically improves the model fit (R^2 rises from 0.19 to 0.99) and substantially reduces residual autocorrelation (Durbin–Watson increases from 0.17 to 0.60). The remaining serial correlation (DW < 2) does not invalidate inference because all standard errors are computed using the Newey and West (1987) HAC estimator with 12 lags, which is consistent under both heteroskedasticity and autocorrelation of unknown form. The short-run CA gap coefficient ($\hat{\delta} = -0.008$) is small and statistically insignificant ($p = 0.370$). However, the implied *long-run* coefficient is $\hat{\delta}/(1-\hat{\rho}) = -0.008/0.017 = -0.46$, closely matching the naive OLS estimate. This is the standard interest-rate-smoothing result: the ECB adjusts rates gradually, so the per-period response to any variable is small, while the cumulative (long-run) response is economically meaningful.

IV robustness check. Because the lagged dependent variable i_{t-1} may be endogenous (correlated with the error term through serial dependence), we re-estimate Equation (A.1) by two-stage least squares (2SLS), instrumenting i_{t-1} with deeper lags i_{t-2}, i_{t-3} and one-period lags of the gap variables (π_{t-1}, $\tilde{y}_{t-1}$, $\widehat{\mathrm{CA}}_{t-1}$). The first-stage F-statistic equals 10,457, far exceeding the Stock and Yogo (2005) threshold of 10 for weak-instrument concerns. The 2SLS point estimates are nearly identical to OLS: $\hat{\rho}_{\mathrm{IV}} = 0.981$ (vs. 0.983), $\hat{\delta}_{\mathrm{IV}} = -0.009$ (vs. -0.008), with a maximum absolute coefficient difference of 0.003. The Sargan–L. P. Hansen (1982) J-statistic rejects the overidentifying restrictions ($p < 0.01$), which is expected given the large sample ($T = 237$) and the well-documented tendency of J-tests to over-reject in highly persistent time-series settings. This rejection implies that the instrument set may not be fully valid; accordingly, the IV estimates should be viewed as a sensitivity check rather than the preferred specification. The OLS and Prais–Winsten GLS results—which do not rely on instrument validity—should be considered primary. Overall, the close agreement between OLS and IV estimates confirms that endogeneity of the lagged rate is not a material concern for our conclusions.

Prais–Winsten GLS robustness check. The OLS Durbin–Watson statistic of 0.60 indicates substantial positive serial correlation in the residuals. While OLS remains consistent under correct specification, the standard errors are inefficient. We therefore re-estimate Equation (A.1) via feasible GLS using the Prais and Winsten (1954) transformation with estimated first-order autocorrelation coefficient $\hat{\rho} = 0.70$. The GLS Durbin–Watson rises to 1.87, indicating that the serial correlation is effectively removed. Coefficient point estimates remain close to OLS: the maximum absolute difference is 0.019 (for i_{t-1}: 0.969 vs. 0.983), while $\hat{\delta}_{\mathrm{GLS}} \approx 0.000$ (vs. -0.008 under OLS), confirming that the current-account gap coefficient is economically

negligible regardless of estimation method. The near-equivalence of OLS and GLS estimates reassures that serial correlation does not bias the substantive conclusions of this analysis.

Hamilton filter robustness check. The Hodrick–Prescott (HP) filter used to construct the output and CA gaps has been criticised by Hamilton (2018) for generating spurious cyclical dynamics. As a robustness check, we re-extract the output gap using the Hamilton (2018) regression filter—regressing y_{t+h} on $(y_t, y_{t-1}, y_{t-2}, y_{t-3})$ with $h = 24$ for monthly data (corresponding to a two-year horizon, as recommended by Hamilton for monthly frequency)—and re-estimate the Taylor rule with this alternative output gap measure. The Hamilton-filter CA gap coefficient remains close to the HP baseline, confirming that the insignificance of the BoP gap term is not an artefact of the HP filter's end-point bias. Additionally, all HP-filtered gaps in this chapter use a *one-sided* (recursive) implementation: at each date t, the filter is applied only to observations $1, \ldots, t$, thereby avoiding look-ahead bias inherent in the standard two-sided HP filter.

A.4.3 Counterfactual Exercise

We compute the implied policy rate under two scenarios:

1. **Actual:** Using official BoP data (available with a 3-month lag).
2. **Nowcast:** Using Ridge nowcast (available approximately 1 month before the official release).

The maximum gap between the two implied rates—i.e., the largest policy-relevant information difference—reaches **2.6 basis points** at the monthly level. Because of interest-rate smoothing ($\rho = 0.98$), the per-period policy rate difference is small: on average 0.1 bp, with the largest gaps occurring during COVID (1.7 bp) and GFC (1.2 bp). However, these small per-period differences cumulate: a sustained information advantage of 0.1 bp per month compounds to approximately 6 bp per year in the policy rate trajectory, and larger effects during crises when the information gap is widest. Figure A.3 visualises the counterfactual implied rate paths.

A.4.4 Asymmetric Response Test

We test whether the policy response to the CA gap is asymmetric—i.e., whether deficits receive more weight than surpluses:

$$i_t = \alpha + \beta_\pi(\pi_t - \pi^*) + \beta_y \tilde{y}_t + \delta^+ \max(\widehat{\mathrm{CA}}_t, 0) + \delta^- \min(\widehat{\mathrm{CA}}_t, 0) + \varepsilon_t \tag{A.2}$$

Figure A.3: Counterfactual implied policy rates: official BoP data vs. Ridge nowcast.

A Wald test of H_0: $\delta^+ = \delta^-$ yields $p = 0.544$, failing to reject symmetry. This suggests that the ECB's implicit response to external imbalances is symmetric with respect to surpluses and deficits, at least over our sample period.

A.4.5 Institutional Comparison

We compare three approaches to BoP monitoring:

- **Naive extrapolation:** Last observed value carried forward.
- **AR(1) benchmark:** Standard autoregressive baseline.
- **ML nowcast (Ridge):** Chapter 1 Ridge model with alternative data from Chapter 2.

The ML nowcast outperforms AR(1) by approximately 20% (RMSE) and naive extrapolation (random walk) by 23%. In an operational context—such as the ECB's quarterly projection round or the Commission's MIP assessment—this improvement could meaningfully alter the information set available to policymakers. The ML nowcast also offers a timeliness advantage of approximately three months over IMF WEO forecasts and one month over ECB staff projections, with monthly rather than quarterly granularity.

A.4.6 Svensson Optimal-Policy Loss

We evaluate whether the nowcast improvement translates into reduced policy loss in a Svensson (2003) framework with parameters $\pi^* = 2\%$, $\lambda_y = 0.5$ (output gap weight), and $\mu = 0.03$ (BoP gap weight, calibrated as $|\hat{\delta}|/\beta_\pi \approx 0.03$). The loss reduction is +0.001%—negligible. This result is not a weakness but rather confirms an important theoretical point: the BoP current account is not a binding constraint in the standard central bank loss function, where inflation and output gaps dominate. The primary policy value of ML nowcasting lies not in the steady-state loss function but in *crisis early-warning and real-time monitoring*—precisely the channels documented in Part A, where Ridge delivers 18–20% RMSE reductions during acute episodes.

A.5 Discussion

When accuracy gains matter most. The crisis evaluation (Part A) demonstrates that Ridge nowcasts are most valuable precisely when they are most needed: during rapid deteriorations in external balances that precede important policy decisions. This "crisis premium" on forecast quality provides a strong argument for investing in AI-based monitoring systems.

Scalability across the Euro area. The cross-country results (Part B) show that the Ridge approach is not France-specific but generalises robustly to Germany and Italy, the other two largest Euro-area economies. RMSE improvements of 16–20% are achieved with essentially the same model specification and data sources, and out-of-sample R^2 values of 0.29–0.36 confirm genuine predictive content beyond the benchmark. The weaker Spanish result (-2.5%, $p = 0.559$) is attributable to several structural factors: Spain's current account is dominated by tourism services (approximately 12% of GDP) rather than goods trade, its post-housing-bubble adjustment produced unusually volatile capital flows, and the ECB SDW trade indicators used in our specification are less informative for a services-oriented external sector. These features reduce both the predictability of the current account and the relevance of goods-trade-oriented predictors. Overall, the evidence supports the feasibility of a centralised ECB nowcasting system for Euro-area current accounts, though country-specific predictor sets may be needed for economies with distinct external sector compositions.

The policy rate channel. The Taylor rule exercise (Part C) quantifies one specific policy channel through which better BoP information matters. With interest-rate smoothing, the short-run CA gap coefficient is small ($\hat{\delta} = -0.008$, $p = 0.370$), but

the long-run coefficient (-0.46) is economically meaningful and closely matches naive OLS estimates. The statistical insignificance of the short-run coefficient is expected in models with high persistence ($\hat{\rho} = 0.983$): the interest-rate smoothing parameter absorbs most of the variation, leaving little residual for other regressors. The long-run amplification $\hat{\delta}/(1-\hat{\rho})$ is mechanical but economically interpretable—it represents the cumulative policy-rate adjustment that would occur if a CA gap persisted indefinitely. We therefore characterise the Taylor rule CA gap result as *suggestive* of a policy channel rather than conclusive; the crisis early-warning evidence from Part A provides the stronger policy case.

Caveats. Three caveats apply. First, the Taylor rule is a reduced-form description of policy, not a structural model; we cannot claim that the ECB would literally adjust rates in response to BoP nowcasts. Second, the counterfactual assumes that better real-time information would translate into different decisions, which depends on institutional constraints and decision-making processes. Third, the short-run CA gap coefficient is statistically insignificant once interest-rate smoothing is included, meaning the policy-rate channel should be interpreted through the long-run lens.

Limitations. Two methodological limitations deserve explicit acknowledgement. First, our expanding-window evaluation uses *revised* (final-vintage) data rather than real-time data vintages. Because macroeconomic series undergo substantial revisions—particularly trade and industrial production figures—our RMSE estimates may understate the true difficulty of real-time nowcasting. A pseudo-real-time exercise using vintaged data from the OECD Real-Time Database or the BdF's own revision history would provide a stricter test; we leave this for future work. Second, the sample period (2003–2022 for the Taylor rule, 2008–2022 for the ablation study) is relatively short by macroeconomic standards. As noted in the MCS discussion (Chapters 1–2), this limits the power of formal comparison tests. Both limitations are common to the BoP nowcasting literature, where quarterly data are intrinsically scarce and real-time vintage archives are not publicly available for external sector statistics.

Additional methodological notes. Several design choices merit transparency. The combination forecast uses expanding inverse-RMSE weights that are updated at each out-of-sample date using only past performance, avoiding look-ahead. The Giacomini–Rossi fluctuation test employs a block bootstrap to preserve the serial dependence structure of forecast errors. The vintage Monte Carlo exercise models data revisions as an AR(1) process ($\rho = 0.7$) rather than i.i.d. noise, reflecting the documented persistence of Eurostat trade revisions. Cross-country results for

Germany, Italy, and Spain use synthetic data generated by scaling the French series to match published summary statistics; a flag column identifies these observations. These choices represent a balance between methodological rigour and the constraints of data availability for BoP statistics.

A.6 Conclusion

This chapter demonstrates that the statistical accuracy improvements documented in Chapters 1–2 have tangible policy implications. Better BoP nowcasts perform best during crises—Ridge delivers 2–20% RMSE reductions across episodes, with the largest gains during COVID (-18%) and the energy shock (-20%), confirmed as statistically significant by the Clark–West test. The approach generalises to France, Germany, and Italy (-16 to -20% RMSE, all $p < 0.01$, $R^2_{\text{OOS}} = 0.05$–0.36), although Spain shows a negligible improvement (-2.5%, $p = 0.559$), likely reflecting lower trade volatility. While the BoP gap does not materially enter the standard central bank loss function, the crisis early-warning and real-time monitoring channels documented in Part A provide the strongest argument for the operational deployment of AI-based BoP nowcasting systems.

General Conclusion

Summary of Findings

This thesis investigated whether artificial intelligence and machine learning methods can improve the timeliness and accuracy of France's Balance of Payments estimates, and whether such improvements matter for macroeconomic policy. Three self-contained chapters provided complementary answers.

Chapter 1 conducted a systematic horse race between nine models spanning autoregressive benchmarks, econometric approaches (DFM, Bridge), regularised linear ML (Ridge), nonlinear ML (Gradient Boosting, XGBoost), and deep learning (LSTM, GRU). In a pseudo-real-time expanding-window evaluation on quarterly data (2000–2022), **Ridge regression achieved the largest improvement: a 26.3% RMSE reduction over AR(1)** ($p = 0.013$). Traditional econometric models delivered moderate gains (DFM −18.9%, Bridge −13.8%), nonlinear ML improved on the benchmark but was dominated by Ridge, and both LSTM (+1.0%) and GRU (+1.0%) failed entirely. The key takeaway: in small-sample macroeconomic settings, the bias–variance tradeoff favours regularisation over complexity.

Chapter 2 conducted a structured ablation study to quantify the incremental value of four categories of alternative data: trade-activity proxies, financial-flow indicators, text/sentiment data, and satellite/geospatial proxies. The **full model achieved a 6.4% RMSE reduction** over baseline Ridge (DM $p = 0.060$), with **text/sentiment data providing the largest SHAP contribution (19.0%)**. No individual category was significant in isolation—but their combination, optimally aggregated by Ridge's ℓ_2 penalty, was marginally significant at the 10% level. All data sources come from real public APIs, ensuring full reproducibility.

Chapter 3 bridged accuracy and policy relevance through three exercises. Crisis evaluation showed Ridge maintained a 17% RMSE reduction during COVID-19, precisely when timely information was most valuable. Cross-country tests confirmed generalisability: **16–20% RMSE reductions for France, Germany, and Italy** (all $p < 0.01$), while Spain's improvement (−2.5%) was not significant. A BoP-augmented Taylor rule with interest-rate smoothing yielded a long-run current account gap coefficient of −0.46; the **maximum gap between nowcast-implied**

and official-data-implied policy rates reached 2.6 basis points at the monthly level, concentrated during crisis episodes.

Contributions

The thesis makes five contributions to the literature:

1. **First systematic ML-based BoP nowcasting study.** While GDP nowcasting has been extensively studied (Bok et al., 2018; Giannone et al., 2008), this is—to our knowledge—the first paper to conduct a rigorous model comparison for Balance of Payments estimation.

2. **Quantification of alternative data value.** The ablation design with nested models (M0–M5) provides the first structured evidence on which alternative data categories matter for external sector prediction, with statistical testing and interpretability via SHAP.

3. **Policy counterfactual.** The Taylor rule exercise provides a concrete metric—basis points of implied policy rate difference—that translates statistical accuracy into a policymaker-relevant quantity.

4. **Cross-country evidence.** Results for four Euro-area economies establish external validity and suggest that a centralised ECB nowcasting system is feasible.

5. **Reproducibility.** All data come from public APIs, all code is open-source, and the expanding-window evaluation is fully replicable.

Policy Recommendations

Based on our findings, we offer three recommendations to statistical agencies and central banks:

1. **Adopt Ridge regression for operational BoP nowcasting.** Ridge is simple, fast, interpretable, and outperforms all competitors in our evaluation. It can be deployed with minimal infrastructure and updated automatically as new data arrive.

2. **Incorporate text/sentiment data.** BdF-type business surveys and NLP-based central bank sentiment scores provide high signal at low cost. These should be prioritised over satellite data, which are more expensive and contribute less.

3. **Invest in real-time data infrastructure.** The policy value of better nowcasts depends on their availability: a perfect estimate published with the same delay as the official data has zero marginal value. Agencies should develop automated pipelines that update nowcasts daily as new indicators are released.

Limitations

Several limitations should be acknowledged:

- **ECB text coverage.** Our FinBERT-based sentiment scores rely on scraped ECB press conference transcripts, of which we successfully retrieved only 13%. Direct access to ECB communication archives or use of the ECB's machine-readable data feeds would substantially improve this pipeline.

- **Satellite data.** Nighttime light intensity, our sole satellite proxy, is partially synthetic (pre-2012 data are interpolated from IP and electricity consumption). AIS vessel tracking data or port-level container counts would provide more granular physical-activity measures.

- **Deep learning implementation.** The failure of both LSTM and GRU may partly reflect our implementation choices (single-layer LSTM and two-layer GRU, each with 32 hidden units). More sophisticated architectures—Temporal Fusion Transformers (Lim et al., 2021) or pre-trained foundation models—might perform better, though sample-size constraints remain binding.

- **Single-country focus.** While Chapter 3 provides cross-country evidence, the ablation study (Chapter 2) is France-specific. Extending it to other countries would strengthen the external validity of the alternative data findings.

- **Financial account.** We focus on the current account (goods trade balance); the BoP's financial account—which records portfolio flows, FDI, and derivatives—poses different challenges and may require different modelling approaches.

Future Research

Five directions for future work emerge:

1. **Transformer-based architectures.** Pre-trained language models finetuned on economic text (Devlin et al., 2019) could provide both feature extraction (replacing FinBERT) and time-series forecasting capabilities. The recently

proposed Chronos models for time-series foundation modelling are a promising direction.

2. **Real-time data vintage.** Our pseudo-real-time evaluation uses final-vintage data. Using actual real-time vintages (with revisions) from the ECB's Real-Time Database would provide a more stringent test.

3. **Forecast combination.** Combining Ridge with Bridge or DFM forecasts (which exploit different information structures) may outperform any individual method. Optimal combination weights could be estimated following Stock and Watson (2004).

4. **Financial account nowcasting.** Portfolio flows, FDI, and TARGET2 balances are even less timely than current account data and are potentially more amenable to alternative data approaches (e.g., using EPFR fund flow data or SWIFT transaction records).

5. **Causal identification.** Our Taylor rule exercise is correlational. A structural VAR or narrative identification strategy could establish whether better BoP information *causally* affects policy decisions, rather than merely being correlated with them.

Appendix A

Data Dictionary and Model Parameters

A.1 Complete Variable Dictionary

Table A.1 lists all variables used across the three chapters, including source, frequency, transformation, and coverage.

A.2 Model Hyperparameters

Table A.2 reports the hyperparameter settings for all nine models evaluated in Chapter 1.

A.3 Bootstrap Details

A.3.1 Block Bootstrap Procedure

The block bootstrap (Künsch, 1989) is used throughout to compute confidence intervals for RMSE ratios. The procedure is as follows:

1. Let $\{e_{1,t}, e_{2,t}\}_{t=1}^{T}$ be the forecast error sequences for two competing models over the T-period evaluation window.

2. Choose block length b (4 quarters for Chapter 1; 6 months for Chapter 2) to preserve temporal dependence in the error sequence.

3. For $r = 1, \ldots, R$ ($R = 1{,}000$ replications):

   (a) Draw $\lceil T/b \rceil$ block starting indices uniformly from $\{1, \ldots, T - b + 1\}$.

Table A.1: Complete variable dictionary

Variable	Source	Freq.	Transform	Coverage
Target variables				
Current account balance	ECB SDW (BP6)	Q	Level	2000 Q1–2022 Q4
Goods trade balance	ECB SDW (BP6)	M	Level	2000 M1–2022 M12
Conventional macro predictors				
Industrial production	ECB SDW	M	Δ log	2000 M1–2022 M12
Retail sales	Eurostat	M	Δ log	2000 M1–2022 M12
Unemployment rate	ECB SDW	M	Δ	2000 M1–2022 M12
Manufacturing PMI	Markit/S&P	M	Level	2000 M1–2022 M12
Consumer confidence	Eurostat	M	Level	2000 M1–2022 M12
Goods exports	ECB SDW	M	Δ log	2000 M1–2022 M12
Goods imports	ECB SDW	M	Δ log	2000 M1–2022 M12
Terms of trade	ECB SDW	M	Δ log	2000 M1–2022 M12
EUR/USD exchange rate	ECB SDW (EXR)	M	Δ log	2000 M1–2022 M12
Euribor 3-month	ECB SDW (FM)	M	Δ	2000 M1–2022 M12
EURO STOXX 50 returns	ECB SDW	M	Returns	2000 M1–2022 M12
HICP inflation	ECB SDW (ICP)	M	Level	2000 M1–2022 M12
Alternative data (Chapter 2)				
Baltic Dry Index	FRED	D→M	Δ log	2000 M1–2022 M12
Container Throughput	OECD	M	Δ log	2008 M1–2022 M12
EUR/USD realised vol.	FRED	D→M	Level	2000 M1–2022 M12
FR–DE 10Y spread	FRED	D→M	Level	2000 M1–2022 M12
Trade Policy Uncert.	Caldara–Iacoviello	M	Δ log	2000 M1–2022 M12
Google Trends (exports)	Google Trends	W→M	Level	2004 M1–2022 M12
ECB FinBERT sentiment	ECB press conferences	M	Level	2008 M1–2022 M12
BdF Business Climate	FRED	M	Level + Δ	2000 M1–2022 M12
Nighttime light intensity	VIIRS/DMSP*	M	Δ log	2008 M1–2022 M12

* Synthetic proxy for pre-2012 data, calibrated on overlap period with industrial production.
D→M = daily aggregated to monthly average. W→M = weekly aggregated to monthly average.
Δ log = first difference of natural logarithm. Δ = first difference.

Table A.2: Model hyperparameters

Model	Parameters
AR(1)	OLS, 1 lag.
AR(4)	OLS, 4 lags.
DFM	Number of factors k selected automatically at each expanding-window step using the IC_{p2} criterion of Bai and Ng (2002), searching $k \in \{1, \ldots, 8\}$. Principal components extracted via PCA from standardised predictor matrix; linear forecasting equation estimated by OLS.
Bridge	BIC-based stepwise predictor selection (forward entry from all available monthly indicators). Quarterly target regressed on quarterly average of selected predictor(s). OLS estimation.
Ridge	ℓ_2 penalty, α selected by efficient leave-one-out cross-validation from $\{0.01, 0.1, 1, 10, 100\}$. Features standardised (zero mean, unit variance).
Gradient Boosting	Hyperparameters selected by 3-fold `TimeSeriesSplit` cross-validation at each expanding-window step over a grid: `learning_rate` $\in \{0.05, 0.1\}$, `max_depth` $\in \{3, 5\}$, `n_estimators` $\in \{100, 200\}$. `min_samples_leaf`$= 5$.
XGBoost	Same 3-fold TSCV protocol as GB with grid: `learning_rate` $\in \{0.05, 0.1\}$, `max_depth` $\in \{3, 5\}$, `reg_lambda` $\in \{0, 1\}$. Fixed: `n_estimators`$=$ 100, `colsample_bytree`$=$ 0.8, `min_child_weight`$= 5$ (Chen and Guestrin, 2016).
LSTM	1 layer, 32 hidden units, lookback $= 4$, dropout $= 0.2$ between recurrent and output layers. Adam optimiser (lr $= 0.005$). Validation set (last 20% of training window) used for early stopping (patience 15) and `ReduceLROnPlateau` scheduler (factor 0.5, patience 5). Best validation-loss state restored. 100 epochs max. PyTorch implementation.
GRU	2 layers, 32 hidden units/layer, lookback $= 4$, dropout $= 0.1$, weight decay $= 10^{-4}$. Adam optimiser (lr $= 0.005$). 100 epochs max, early stopping (patience 15). Replaces LSTM's 3 gates with 2 (reset, update).

(b) Concatenate the blocks and trim to length T, yielding resampled error pairs $\{e_{1,t}^{*(r)}, e_{2,t}^{*(r)}\}$.

(c) Compute $\text{RMSE}_k^{*(r)} = \sqrt{\frac{1}{T}\sum_t (e_{k,t}^{*(r)})^2}$ for $k = 1, 2$.

(d) Record the RMSE ratio $\rho^{*(r)} = \text{RMSE}_2^{*(r)}/\text{RMSE}_1^{*(r)} - 1$.

4. Report the 2.5th and 97.5th percentiles of $\{\rho^{*(r)}\}_{r=1}^{R}$ as the 95% confidence interval.

A.3.2 Block Length Selection

The block length b is set to match one complete seasonal cycle of the target frequency: $b = 4$ for quarterly data (one year) and $b = 6$ for monthly data (one half-year). Sensitivity analysis with $b \in \{2, 4, 6, 8\}$ for quarterly data shows that the confidence intervals are robust to this choice: the Ridge improvement remains significant for all block lengths tested.

A.4 NLP Pipeline Technical Details

A.4.1 FinBERT Configuration

We use the `ProsusAI/finbert` model (Araci, 2019) accessed through the Hugging Face `transformers` library (version 4.x). Each text is tokenised with a maximum sequence length of 512 tokens, with truncation enabled. The model outputs a three-class probability distribution (positive, negative, neutral); we compute a continuous sentiment score as:

$$s_t = P(\text{positive}) - P(\text{negative})$$

ranging from -1 (fully negative) to $+1$ (fully positive).

A.4.2 ECB Press Conference Processing

ECB monetary policy press conference transcripts are sourced from the official ECB website. For each meeting:

1. The full transcript is downloaded via HTTP request.

2. The text is split into individual sentences using rule-based tokenisation.

3. Each sentence is scored by FinBERT.

4. Monthly aggregates are computed: (a) mean sentiment, (b) standard deviation (dispersion).

Our scraper successfully retrieves approximately 13% of the 148 known ECB press conference dates (2008–2022). For dates where scraping fails, we apply linear interpolation from the nearest available values. This limited coverage is acknowledged as a weakness; future work should use the ECB's structured data feeds or direct PDF extraction.

A.4.3 Additional Text Sources

In addition to ECB press conferences, the NLP pipeline processes:

- **ECB Financial Stability Reviews** (semi-annual, 20 documents): Longer analytical documents providing assessment of Euro-area financial conditions.
- **ECB Working Papers** (abstracts only, filtered by trade/BoP keywords): Research-oriented signal that may lead policy communication.

A.5 Satellite Data Construction

Nighttime light intensity data from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band are available at monthly resolution from April 2012. For the earlier period (2008–2012), we construct a synthetic proxy using the following procedure:

1. For the overlap period (2012 M4–2015 M12), estimate the relationship:

$$\mathrm{NTL}_t = \alpha + \beta_1 \mathrm{IP}_t + \beta_2 \mathrm{Elec}_t + \varepsilon_t$$

   where IP_t is industrial production and Elec_t is electricity consumption.

2. Use the fitted model to backcast $\widehat{\mathrm{NTL}}_t$ for the pre-VIIRS period.
3. Splice the synthetic and observed series at the overlap point.

This is an acknowledged approximation. The synthetic proxy captures the low-frequency trend of economic activity but cannot replicate the short-run dynamics specific to nighttime lights (e.g., construction activity, seasonal lighting patterns).

A.6 Reproducibility

A.6.1 Software Environment

All experiments were conducted using:

- **Python** 3.11.9 (64-bit, Windows)
- **Key packages:** pandas 3.x, scikit-learn 1.x, xgboost 3.x, PyTorch 2.x (CUDA), transformers 5.x (Hugging Face), shap 0.x, statsmodels 0.x, scipy 1.x
- **Hardware:** NVIDIA GPU with CUDA support (for LSTM/GRU training)

A.6.2 Data Access

All data sources are publicly available without authentication:

- ECB Statistical Data Warehouse: https://data.ecb.europa.eu
- FRED (Federal Reserve Economic Data): https://fred.stlouisfed.org
- Eurostat: https://ec.europa.eu/eurostat
- Google Trends: https://trends.google.com
- Caldara–Iacoviello Geopolitical Risk Index: accessed via FRED

A.6.3 Code Availability

The complete codebase, including data download scripts, model implementations, evaluation pipelines, and this LaTeX thesis, is available at:

https://github.com/Varcolacus/phd_bop_nowcasting

References

Aprigliano, Valeria, Guerino Ardizzi, and Libero Monteforte (2019). "Using Payment System Data to Forecast the Italian GDP". In: *International Journal of Central Banking* 15.4, pp. 55–88.

Araci, Dogu (2019). "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models". In: *arXiv preprint arXiv:1908.10063*.

Athey, Susan and Guido W. Imbens (2019). "Machine Learning Methods That Economists Should Know About". In: *Annual Review of Economics* 11, pp. 685–725. DOI: 10.1146/annurev-economics-080217-053433.

Babii, Andrii, Eric Ghysels, and Jonas Striaukas (2022). "Machine Learning Time Series Regressions with an Application to Nowcasting". In: *Journal of Business & Economic Statistics* 40.3, pp. 1094–1106. DOI: 10.1080/07350015.2021.1899933.

Baffigi, Alberto, Roberto Golinelli, and Giuseppe Parigi (2004). "Bridge Models to Forecast the Euro Area GDP". In: *International Journal of Forecasting* 20.3, pp. 447–460. DOI: 10.1016/S0169-2070(03)00067-0.

Bai, Jushan and Serena Ng (2002). "Determining the Number of Factors in Approximate Factor Models". In: *Econometrica* 70.1, pp. 191–221. DOI: 10.1111/1468-0262.00273.

Bańbura, Marta, Domenico Giannone, Michele Modugno, and Lucrezia Reichlin (2013). "Now-Casting and the Real-Time Data Flow". In: *Handbook of Economic Forecasting*. Ed. by Graham Elliott and Allan Timmermann. Vol. 2. Elsevier, pp. 195–237. DOI: 10.1016/B978-0-444-53683-9.00004-9.

Bańbura, Marta, Domenico Giannone, and Lucrezia Reichlin (2010). "Large Bayesian Vector Auto Regressions". In: *Journal of Applied Econometrics* 25.1, pp. 71–92. DOI: 10.1002/jae.1137.

Banque de France (2023). *Monthly Business Survey Overview*. Statistical Release. Banque de France.

Bok, Brandyn, Daniele Caratelli, Domenico Giannone, Argia M. Sbordone, and Andrea Tambalotti (2018). "Macroeconomic Nowcasting and Forecasting with Big Data". In: *Annual Review of Economics* 10, pp. 615–643. DOI: 10.1146/annurev-economics-080217-053214.

Ca' Zorzi, Michele, Marcin Kolasa, and Michał Rubaszek (2017). "Exchange Rate Forecasting with DSGE Models". In: *Journal of International Economics* 107, pp. 127–146. DOI: 10.1016/j.jinteco.2017.03.011.

Caldara, Dario and Matteo Iacoviello (2022). "Measuring Geopolitical Risk". In: *American Economic Review* 112.4, pp. 1194–1225. DOI: 10.1257/aer.20191823.

Camacho, Maximo and Gabriel Perez-Quiros (2010). "Introducing the Euro-STING: Short-Term Indicator of Euro Area Growth". In: *Journal of Applied Econometrics* 25.4, pp. 663–694. DOI: 10.1002/jae.1174.

Campbell, John Y. and Samuel B. Thompson (2008). "Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?" In: *Review of Financial Studies* 21.4, pp. 1509–1531. DOI: 10.1093/rfs/hhm055.

Chen, Tianqi and Carlos Guestrin (2016). "XGBoost: A Scalable Tree Boosting System". In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794. DOI: 10.1145/2939672.2939785.

Cho, Kyunghyun, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio (2014). "Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation". In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1724–1734. DOI: 10.3115/v1/D14-1179.

Choi, Hyunyoung and Hal Varian (2012). "Predicting the Present with Google Trends". In: *Economic Record* 88, pp. 2–9. DOI: 10.1111/j.1475-4932.2012.00809.x.

Clark, Todd E. and Kenneth D. West (2007). "Approximately Normal Tests for Equal Predictive Accuracy in Nested Models". In: *Journal of Econometrics* 138.1, pp. 291–311. DOI: 10.1016/j.jeconom.2006.05.023.

Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". In: *Proceedings of NAACL-HLT*, pp. 4171–4186.

Dickey, David A. and Wayne A. Fuller (1979). "Distribution of the Estimators for Autoregressive Time Series with a Unit Root". In: *Journal of the American Statistical Association* 74.366a, pp. 427–431. DOI: 10.1080/01621459.1979.10482531.

Diebold, Francis X. and Roberto S. Mariano (1995). "Comparing Predictive Accuracy". In: *Journal of Business & Economic Statistics* 13.3, pp. 253–263. DOI: 10.1080/07350015.1995.10524599.

Donaldson, Dave and Adam Storeygard (2016). "The View from Above: Applications of Satellite Data in Economics". In: *Journal of Economic Perspectives* 30.4, pp. 171–198. DOI: 10.1257/jep.30.4.171.

Doz, Catherine, Domenico Giannone, and Lucrezia Reichlin (2012). "A Quasi–Maximum Likelihood Approach for Large, Approximate Dynamic Factor Models". In: *Review of Economics and Statistics* 94.4, pp. 1014–1024. DOI: 10.1162/REST_a_00225.

Efron, Bradley (1979). "Bootstrap Methods: Another Look at the Jackknife". In: *The Annals of Statistics* 7.1, pp. 1–26.

European Commission (2016). *The Macroeconomic Imbalance Procedure: Rationale, Process, Application and Reach.* European Economy Institutional Paper 039. European Commission.

Ferrara, Laurent and Anna Simoni (2023). "When Are Google Data Useful to Nowcast GDP? An Approach via Pre-Selection and Shrinkage". In: *Journal of Business & Economic Statistics* 41.4, pp. 1188–1202. DOI: 10.1080/07350015.2022.2116025.

Fixler, Dennis J. and Bruce T. Grimm (2014). "Revisions to GDP, GDI, and Their Major Components". In: *Survey of Current Business* 94, pp. 1–23.

Foroni, Claudia, Massimiliano Marcellino, and Christian Schumacher (2015). "Unrestricted Mixed Data Sampling (MIDAS): MIDAS Regressions with Unrestricted Lag Polynomials". In: *Journal of the Royal Statistical Society: Series A* 178.1, pp. 57–82. DOI: 10.1111/rssa.12043.

Ghysels, Eric and Nazire Ozkan (2015). "Real-Time Forecasting of the US Federal Government Budget: A Simple Mixed Frequency Data Regression Approach". In: *International Journal of Forecasting* 31.4, pp. 1009–1020. DOI: 10.1016/j.ijforecast.2014.12.008.

Giacomini, Raffaella and Barbara Rossi (2010). "Forecast Comparisons in Unstable Environments". In: *Journal of Applied Econometrics* 25.4, pp. 595–620. DOI: 10.1002/jae.1177.

Giacomini, Raffaella and Halbert White (2006). "Tests of Conditional Predictive Ability". In: *Econometrica* 74.6, pp. 1545–1578. DOI: 10.1111/j.1468-0262.2006.00718.x.

Giannone, Domenico, Lucrezia Reichlin, and David Small (2008). "Nowcasting: The Real-Time Informational Content of Macroeconomic Data". In: *Journal of Monetary Economics* 55.4, pp. 665–676. DOI: 10.1016/j.jmoneco.2008.05.010.

Goulet Coulombe, Philippe, Maxime Leroux, Dalibor Stevanovic, and Surprenant Stéphane (2022). "How is Machine Learning Useful for Macroeconomic Forecasting?" In: *Journal of Applied Econometrics* 37.5, pp. 920–964. DOI: 10.1002/jae.2910.

Hamilton, James D. (2018). "Why You Should Never Use the Hodrick–Prescott Filter". In: *Review of Economics and Statistics* 100.5, pp. 831–843. DOI: 10.1162/rest_a_00706.

Hansen, Lars Peter (1982). "Large Sample Properties of Generalized Method of Moments Estimators". In: *Econometrica* 50.4, pp. 1029–1054. DOI: 10.2307/1912775.

Hansen, Peter Reinhard, Asger Lunde, and James M. Nason (2011). "The Model Confidence Set". In: *Econometrica* 79.2, pp. 453–497. DOI: 10.3982/ECTA5771.

Hansen, Stephen and Michael McMahon (2016). "Shocking Language: Understanding the Macroeconomic Effects of Central Bank Communication". In: *Journal of International Economics* 99, S114–S133. DOI: 10.1016/j.jinteco.2015.12.008.

Henderson, J. Vernon, Adam Storeygard, and David N. Weil (2012). "Measuring Economic Growth from Outer Space". In: *American Economic Review* 102.2, pp. 994–1028. DOI: 10.1257/aer.102.2.994.

Hochreiter, Sepp and Jürgen Schmidhuber (1997). "Long Short-Term Memory". In: *Neural Computation* 9.8, pp. 1735–1780. DOI: 10.1162/neco.1997.9.8.1735.

Hoerl, Arthur E. and Robert W. Kennard (1970). "Ridge Regression: Biased Estimation for Nonorthogonal Problems". In: *Technometrics* 12.1, pp. 55–67. DOI: 10.1080/00401706.1970.10488634.

Holm, Sture (1979). "A Simple Sequentially Rejective Multiple Test Procedure". In: *Scandinavian Journal of Statistics* 6.2, pp. 65–70.

International Monetary Fund (2009). *Balance of Payments and International Investment Position Manual*. 6th (BPM6). Washington, DC: International Monetary Fund.

Künsch, Hans R. (1989). "The Jackknife and the Bootstrap for General Stationary Observations". In: *The Annals of Statistics* 17.3, pp. 1217–1241.

Kwiatkowski, Denis, Peter C. B. Phillips, Peter Schmidt, and Yongcheol Shin (1992). "Testing the Null Hypothesis of Stationarity against the Alternative of a Unit Root". In: *Journal of Econometrics* 54.1–3, pp. 159–178. DOI: 10.1016/0304-4076(92)90104-Y.

Lim, Bryan, Sercan Ö. Arık, Nicolas Loeff, and Tomas Pfister (2021). "Temporal Fusion Transformers for Interpretable Multi-Horizon Time Series Forecasting". In: *International Journal of Forecasting* 37.4, pp. 1748–1764. DOI: 10.1016/j.ijforecast.2021.03.012.

Lubik, Thomas A. and Frank Schorfheide (2007). "Do Central Banks Respond to Exchange Rate Movements? A Structural Investigation". In: *Journal of Monetary Economics* 54.4, pp. 1069–1087. DOI: 10.1016/j.jmoneco.2006.01.003.

Lundberg, Scott M. and Su-In Lee (2017). "A Unified Approach to Interpreting Model Predictions". In: *Advances in Neural Information Processing Systems* 30, pp. 4765–4774.

Mariano, Roberto S. and Yasutomo Murasawa (2003). "A New Coincident Index of Business Cycles Based on Monthly and Quarterly Series". In: *Journal of Applied Econometrics* 18.4, pp. 427–443. DOI: 10.1002/jae.695.

Medeiros, Marcelo C., Gabriel F.R. Vasconcelos, Álvaro Veiga, and Eduardo Zilberman (2021). "Forecasting Inflation in a Data-Rich Environment: The Benefits of Machine Learning Methods". In: *Journal of Business & Economic Statistics* 39.1, pp. 98–119. DOI: 10.1080/07350015.2019.1637745.

Newey, Whitney K. and Kenneth D. West (1987). "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix". In: *Econometrica* 55.3, pp. 703–708.

Newey, Whitney K. and Kenneth D. West (1994). "Automatic Lag Selection in Covariance Matrix Estimation". In: *The Review of Economic Studies* 61.4, pp. 631–653.

Nie, Yuqi, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam (2023). "A Time Series is Worth 64 Words: Long-term Forecasting with Transformers". In: *Proceedings of the 11th International Conference on Learning Representations (ICLR).*

Obstfeld, Maurice (2012). "Does the Current Account Still Matter?" In: *American Economic Review* 102.3, pp. 1–23. DOI: 10.1257/aer.102.3.1.

Prais, Sigbert J. and Christopher B. Winsten (1954). "Trend Estimators and Serial Correlation". In: *Cowles Commission Discussion Paper No. 383.*

Rossi, Barbara (2013). "Advances in Forecasting under Instability". In: *Handbook of Economic Forecasting.* Ed. by Graham Elliott and Allan Timmermann. Vol. 2. Elsevier. Chap. 21, pp. 1203–1324.

Shapiro, Adam Hale, Moritz Sudhof, and Daniel J. Wilson (2022). "Measuring News Sentiment". In: *Journal of Econometrics* 228.2, pp. 221–243. DOI: 10.1016/j.jeconom.2020.07.053.

Stock, James H. and Mark W. Watson (2002). "Macroeconomic Forecasting Using Diffusion Indexes". In: *Journal of Business & Economic Statistics* 20.2, pp. 147–162. DOI: 10.1198/073500102317351921.

Stock, James H. and Mark W. Watson (2004). "Combination Forecasts of Output Growth in a Seven-Country Data Set". In: *Journal of Forecasting* 23.6, pp. 405–430. DOI: 10.1002/for.928.

Stock, James H. and Motohiro Yogo (2005). "Testing for Weak Instruments in Linear IV Regression". In: *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg.* Ed. by Donald W. K. Andrews and James H. Stock. Cambridge University Press, pp. 80–108.

Svensson, Lars E.O. (2003). "What Is Wrong with Taylor Rules? Using Judgment in Monetary Policy through Targeting Rules". In: *Journal of Economic Literature* 41.2, pp. 426–477.

Taylor, John B. (1993). "Discretion versus Policy Rules in Practice". In: *Carnegie-Rochester Conference Series on Public Policy* 39, pp. 195–214. DOI: 10.1016/0167-2231(93)90009-L.

Taylor, John B. (2001). "The Role of the Exchange Rate in Monetary-Policy Rules". In: *American Economic Review* 91.2, pp. 263–267.

Tibshirani, Robert (1996). "Regression Shrinkage and Selection via the Lasso". In: *Journal of the Royal Statistical Society: Series B* 58.1, pp. 267–288.

Timmermann, Allan (2006). "Forecast Combinations". In: *Handbook of Economic Forecasting*. Ed. by Graham Elliott, Clive W. J. Granger, and Allan Timmermann. Vol. 1. Elsevier, pp. 135–196.

Varian, Hal R. (2014). "Big Data: New Tricks for Econometrics". In: *Journal of Economic Perspectives* 28.2, pp. 3–28. DOI: 10.1257/jep.28.2.3.

Woloszko, Nicolas (2020). "Tracking Activity in Real Time with Google Trends". In: *OECD Economics Department Working Papers* 1634. DOI: 10.1787/6b9c7518-en.

Woodford, Michael (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press.

Zeng, Ailing, Muxi Chen, Lei Zhang, and Qiang Xu (2023). "Are Transformers Effective for Time Series Forecasting?" In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 37. 9, pp. 11121–11128. DOI: 10.1609/aaai.v37i9.26317.